

Компендиум исследований TRIVORTEX

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Abstract

TRIVORTEX — исследовательская программа вокруг задачи трёх тел. Двенадцать сопроводительных исследований — каждое: исполнимый скрипт, коммитнутый JSON-протокол и двуязычная монография — атакуют задачу через двенадцать физических систем, причём в десяти из них в центре стоит лазер. Таблица приводит главное проверенное число каждого исследования; далее следуют досье с полными машинночитаемыми протоколами.

1. Программа одним взглядом

TRIVORTEX — исследовательская программа вокруг задачи трёх тел. Двенадцать сопроводительных исследований — каждое: исполнимый скрипт, коммитнутый JSON-протокол и двуязычная монография — атакуют задачу через двенадцать физических систем, причём в десяти из них в центре стоит лазер. Таблица приводит главное проверенное число каждого исследования; далее следуют досье с полными машинночитаемыми протоколами.

#	Study	Directory	Status
TRX-01	$\mathbb{D} \alpha \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}_{\text{EPH}} \mathbb{D} \frac{3}{4} \mathbb{D}^3 \tilde{\mathbb{N}}_{\text{PAD}} \mathbb{D}^\circ \mathbb{D}$ ${}^2\mathbb{D}, \tilde{\mathbb{N}}_{\text{EPH}} \mathbb{D}^\circ \tilde{\mathbb{N}}_{\text{SFA}} \mathbb{D}, \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2}$ $\mathbb{D} \frac{1}{2} \mathbb{D}^\circ \tilde{\mathbb{N}}_{\text{SS3}}$ $\mathbb{D} \frac{3}{4} \mathbb{D}^3 \tilde{\mathbb{N}}_{\text{PAD}} \mathbb{D}^\circ \mathbb{D} \frac{1}{2} \mathbb{D}, \tilde{\mathbb{N}}_{\text{ESR}} \mathbb{D}$ $\mu \mathbb{D} \frac{1}{2} \mathbb{D} \frac{1}{2} \mathbb{D}^\circ \tilde{\mathbb{N}}_{\text{SS3}} \mathbb{D} \cdot \mathbb{D}^\circ \mathbb{D}$ $\text{'}\mathbb{D}^\circ \tilde{\mathbb{N}}_{\text{ESR}} \mathbb{D}^\circ$ $\tilde{\mathbb{N}}_{\text{EPH}} \tilde{\mathbb{N}}_{\text{PAD}} \tilde{\mathbb{N}}_{\text{PU1}} \tilde{\mathbb{N}}_{\text{NEL}}$	TRX-01-laser-radiation-pressure	PASS

#	Study	Directory	Status
	$\tilde{N}^{\text{EPH}} D_{\mu} D \gg$ $(D \gg D^{\circ} D \cdot D_{\mu} \tilde{N}^{\text{FAD}} D^{\frac{1}{2}} D^{\circ}$ $D_{\ell} \tilde{N}^{\text{FLD}} D \gg D_{\mu} D^2 \tilde{N}^{\text{NEH}} \tilde{N}^{\text{SSZ}}$ $\tilde{N}^{\text{ESR}} D^{\circ} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{EPH}} D_{\mu} \tilde{N}^{\text{SSR}} \tilde{N}^{\text{NEH}})$		
TRX-02	$D D_{\mu} D \cdot D^{\frac{3}{4}} D^{\frac{1}{2}} D^{\circ} D^{\frac{1}{2}} \tilde{N}^{\text{HOF}} D^{\frac{1}{2}} D^{\frac{3}{4}} D_{\mu}$ $\tilde{N}^{\text{EPH}} \tilde{N}^{\text{FAD}} \tilde{N}^{\text{FUL}} \tilde{N}^{\text{HEL}}$ $D^2 D^{\frac{3}{4}} D \gg D^{\frac{1}{2}} D^{\frac{3}{4}} D^2 D^{\frac{3}{4}} D_{\mu}$ $D^2 D \cdot D^{\circ} D, D^{\frac{1}{4}} D^{\frac{3}{4}} D$ $\text{' } D_{\mu} D^1 \tilde{N}^{\text{HOF}} \tilde{N}^{\text{EPH}} D^2 D, D_{\mu}$ $(D^{\text{ST}} \tilde{N}^{\text{RI}} D^{\frac{1}{2}} D \gg D, \hat{a}^{\text{FAD}} \text{' } D$ $D^{\frac{3}{4}} \tilde{N}^{\text{NEH}},$ $\tilde{I}^{\text{ESR}} \hat{a}^{\text{HOF}} \frac{1}{2} \hat{A}^2 \hat{a}^{\text{HOF}} \frac{3}{4} -$ $D^{\frac{3}{4}} D_{\ell} \tilde{N}^{\text{EPH}} D, D^{\circ} D^{\circ})$	TRX-02-three-wave-mixing	PASS
TRX-03	$D^{\text{ST}} D^{\frac{3}{4}} D \gg D_{\mu} D^{\circ} \tilde{N}^{\text{NEH}} D \gg D$ $^{\circ} D, D \cdot \tilde{N}^{\text{EPH}} \tilde{N}^{\text{FAD}} \tilde{N}^{\text{FUL}} \tilde{N}^{\text{HEL}}$ $\tilde{N}^{\text{HOF}} D^{\frac{3}{4}} D \gg D, \tilde{N}^{\text{EPH}} D^{\frac{3}{4}} D^{\frac{1}{2}}$ $D^{\frac{3}{4}} D^2 D^2$ $D^2 D^{\frac{3}{4}} D \gg D^{\frac{3}{4}} D^{\circ} D^{\frac{3}{4}} D^{\frac{1}{2}} D$ $\frac{1}{2} D^{\frac{3}{4}} D^{\frac{1}{4}}$ $D \gg D^{\circ} D \cdot D_{\mu} \tilde{N}^{\text{FAD}} D_{\mu} \tilde{N}^{\text{HOF}}$ $\tilde{N}^{\text{HOF}} D, D^{\frac{1}{2}} \tilde{N}^{\text{HEL}}$ $\tilde{N}^{\text{FAD}} D^{\frac{3}{4}} D^{\frac{1}{2}} D, D \cdot D^{\circ} \tilde{N}^{\text{SSR}} D$ $, D_{\mu} D^1 D^{\frac{1}{4}} D^{\frac{3}{4}} D \text{'}$	TRX-03-soliton-molecule	PASS
TRX-04	$D^{\circ} D^{\frac{3}{4}} \tilde{N}^{\text{EPH}} D^{\frac{3}{4}} D^{\frac{1}{2}} D^{\frac{1}{2}} D^{\circ}$ $\tilde{N}^{\text{SSZ}} D^{\text{I}} D, D$ $\text{' } D^{\circ} D^{\frac{3}{4}} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{EPH}} \tilde{N}^{\text{FLU}} \cdot$ $\tilde{N}^{\text{EPH}} \tilde{N}^{\text{FAD}} D,$ $D^{\circ} D_{\mu} \tilde{N}^{\text{FAD}} \tilde{N}^{\text{FAD}} D^{\frac{3}{4}} D^2 \tilde{N}^{\text{HOF}}$ $D^{\circ} D, \tilde{N}^{\text{HEL}}$ $\tilde{N}^{\text{HOF}} D^{\frac{3}{4}} D \gg D, \tilde{N}^{\text{EPH}} D^{\frac{3}{4}} D^{\frac{1}{2}}$ D° $(D_{\ell} \tilde{N}^{\text{FAD}} D^{\frac{3}{4}} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{EPH}} \tilde{N}^{\text{FAD}}$ $D^{\circ} D^{\frac{1}{2}} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{EPH}} D^2 D_{\mu} D^{\frac{1}{2}} D$ $\frac{1}{2} \tilde{N}^{\text{FLD}} D_{\mu}, 2-D)$	TRX-04-photon-fluid	PASS
TRX-05	$D^{\text{PM}} D_{\ell} \tilde{N}^{\text{EPH}} D, \tilde{N}^{\text{ESR}} D_{\mu} \tilde{N}^{\text{HOF}}$ $D^{\circ} D, D_{\mu} D^2 D, \tilde{N}^{\text{HEL}} \tilde{N}^{\text{FAD}} D,$ $D \gg D^{\circ} D \cdot D_{\mu} \tilde{N}^{\text{FAD}} D^{\frac{1}{2}} \tilde{N}^{\text{FLD}} \tilde{N}^{\text{HEL}}$ $D_{\ell} \tilde{N}^{\text{NEH}} \tilde{N}^{\text{ESR}} D^{\circ} D^{\frac{3}{4}} D^2$ $D,$	TRX-05-optical-vortices	PASS

#	Study	Directory	Status
	$D^0 D_{1/2} D^0 D \gg D_{3/4} D^3 D, \tilde{N}_{SS}$ $\tilde{N}_{HOP}$ $\tilde{N}_{BPH} D_{3/4} \tilde{N}_{ESR} D_{\mu} \tilde{N}_{ESR} D_{1/2} \tilde{N}$ $\tilde{P}_{LD} D_{1/4} D, D^2 D, \tilde{N}_{HEL}$ $\tilde{N}_{PAD} \tilde{N}_{SS} D_{1/4} D,$		
TRX-06	$D_{DCS} \tilde{N}_{BPH} D_{3/4} D_{1/4}$ $D^3 D_{\mu} D \gg D, \tilde{N}_{SS} D^0 D^0 D^0$ $D^0 \tilde{N}_{HEH} D \gg D_{3/4} D_{1/2} D_{3/4} D^2 \tilde{N}$ $\tilde{HOP} D^0 D^0 \tilde{N}_{SS} D \cdot D^0 D$ $D^0 \tilde{N}_{ESR} D^0$ $\tilde{N}_{BPH} \tilde{N}_{PAD} \tilde{N}_{FUI} \tilde{N}_{HEL}$ $\tilde{N}_{BPH} D_{\mu} D \gg$ $(D^0 D \gg D^0 \tilde{N}_{HOP} \tilde{N}_{HOP} D, \tilde{N}_{ESR}$ $D_{\mu} \tilde{N}_{HOP} D^0 D, D^1$ $\tilde{N}_{BPH} \tilde{N}_{PAD} D^0 D_{\mu} D^0 \tilde{N}_{BPH} D_{3/4}$ $\tilde{N}_{PAD} D_{1/2} \tilde{N}_{PLD} D^1$ $D_{1/4} D_{\mu} \tilde{N}_{BPH} D_{3/4} D'$ $D_{ST} D_{3/4} D_{1/2} \tilde{N}_{BPH} D_{\mu}-$ $D_{SCI} D^0 \tilde{N}_{PAD} D \gg D_{3/4})$	TRX-06-helium- three-body	PASS
TRX-07	$D \tilde{N}_{IND} \tilde{N}_{IND} D_{\mu} D^0 \tilde{N}_{BPH}$ $D_{MH} \tilde{N}_{IND} D, D_{1/4} D_{3/4} D^2 D^0:$ $\tilde{N}_{HEH} D_{1/2} D, D^2 D_{\mu} \tilde{N}_{PAD} \tilde{N}_{HOP}$ $D^0 D \gg \tilde{N}_{PLU} D_{1/2} D^0 \tilde{N}_{SS}$ $D^0 D^2 D^0 D_{1/2} \tilde{N}_{BPH} D_{3/4} D^2 D^0$ $\tilde{N}_{SS} \tilde{N}_{IND} D, D \cdot D, D^0 D^0$ $\tilde{N}_{BPH} \tilde{N}_{PAD} \tilde{N}_{FUI} \tilde{N}_{HEL}$ $\tilde{N}_{BPH} D_{\mu} D \gg$	TRX-07-efimov	PASS
TRX-08	$D \tilde{N}_{PAD} D,$ $D \gg D^0 D \cdot D_{\mu} \tilde{N}_{PAD} D_{1/2} D_{3/4}-$ $D_{3/4} \tilde{N}_{HEL} D \gg D^0 D \P D$ $D' \tilde{N}_{FUI} D_{1/2} D_{1/2} \tilde{N}_{PLD} \tilde{N}_{HEL}$ $D, D_{3/4} D_{1/2} D^0 D^2$ $D \gg D, D_{1/2} D_{\mu} D^1 D_{1/2} D_{3/4} D^1$ $D \gg D_{3/4} D^2 \tilde{N}_{HEH} \tilde{N}_{HTS} D^0 D_{\mu}$ $D_{APC} D^0 \tilde{N}_{HEH} D \gg \tilde{N}_{SS}:$ $D_{1/2} D^0 \tilde{N}_{HOP} \tilde{N}_{BPH} D_{3/4} D \gg \tilde{N}_{PLU}$ $D_{1/2} \tilde{N}_{PLD} D^1$ $\tilde{N}_{BPH} \tilde{N}_{PAD} D_{\mu} \tilde{N}_{HEH} D^3 D_{3/4} D \gg$ $\tilde{N}_{PLU} D_{1/2} D, D^0$ $D_{CST} D^0 D^3 \tilde{N}_{PAD} D^0 D_{1/2} D \P D^0$	TRX-08-ion-trap	PASS
TRX-09	$D_{APC} \tilde{N}_{PAD} D_{3/4} D \pm D \gg D_{\mu} D_{1/4}$	TRX-09-vortex-trio	PASS

#	Study	Directory	Status
	$\mathbb{D}^{\circ} \tilde{N}^{\text{BPH}} \tilde{N}^{\text{PAD}} \tilde{N}^{\text{FUE}} \tilde{N}^{\text{HEL}}$ $\mathbb{D}^2 \mathbb{D}, \tilde{N}^{\text{HEL}} \tilde{N}^{\text{PAD}} \mathbb{D} \mu \mathbb{D}^1$ $\mathbb{D}^{\text{SCI}} \mathbb{D}, \tilde{N}^{\text{PAD}} \tilde{N}^{\text{HEL}}$ $\mathbb{D}^3 \mathbb{D} \frac{3}{4} \tilde{N}^{\text{IND}} \mathbb{D}^{\circ} \hat{a}^{\text{PAD}} \mathbb{D} \S \mathbb{D}^{\circ} \mathbb{D}$ $\mathbb{D} \mathbb{D} \gg \tilde{N}^{\text{FLD}} \mathbb{D}^3 \mathbb{D}, \mathbb{D} \frac{1}{2} \mathbb{D}^{\circ}$ $(\mathbb{D}^{\circ} \mathbb{D} \gg \mathbb{D}^{\circ} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{HOF}} \mathbb{D}, \tilde{N}^{\text{ESR}}$ $\mathbb{D} \mu \tilde{N}^{\text{HOF}} \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D}^1$ $\tilde{N}^{\text{SSR}} \mathbb{D}^{\circ} \mathbb{D} \frac{3}{4} \tilde{N}^{\text{PAD}} \tilde{N}^{\text{FLU}})$		
TRX-10	$\mathbb{D}^{\text{SCI}} \mathbb{D} \frac{3}{4} \mathbb{D} \gg \mathbb{D} \mu \mathbb{D} \pm \mathbb{D}^{\circ} \mathbb{D} \frac{1}{2} \mathbb{D}$ $, \tilde{N}^{\text{SSR}}$ $\mathbb{D}^{\text{SCI}} \mathbb{D} \frac{3}{4} \mathbb{D} \cdot \mathbb{D}^{\circ} \mathbb{D}, \hat{a}^{\text{PAD}} \mathbb{D}^{\text{CSI}}$ $\mathbb{D}, \mathbb{D}' \mathbb{D} \frac{3}{4} \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D}^2$ $\mathbb{D}, \mathbb{D} \mu \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \tilde{N}^{\text{PAD}} \tilde{N}^{\text{HEL}}$ $\mathbb{D}, \tilde{N}^{\text{ESR}} \mathbb{D} \mu \tilde{N}^{\text{HOF}} \mathbb{D}^{\circ} \mathbb{D}, \tilde{N}^{\text{HEL}}$ $\tilde{N}^{\text{BPH}} \tilde{N}^{\text{PAD}} \mathbb{D} \frac{3}{4} \mathbb{D}^1 \mathbb{D} \frac{1}{2} \tilde{N}^{\text{FLD}} \tilde{N}^{\text{HEL}}$ $\tilde{N}^{\text{HOF}} \mathbb{D}, \tilde{N}^{\text{HOF}} \tilde{N}^{\text{BPH}} \mathbb{D} \mu \mathbb{D} \frac{1}{4} \mathbb{D}^{\circ}$ $\tilde{N}^{\text{HEL}}$	TRX-10-kozai-lidov	PASS
TRX-11	$\mathbb{D}^{\text{STS}} \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \mathbb{D}^2 \mathbb{D}, \tilde{N}^{\text{BPH}} \mathbb{D}^{\circ} \tilde{N}^{\text{SSR}} \mathbb{D}, \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{1}{2} \tilde{N}^{\text{FLD}} \mathbb{D} \mu$ $\mathbb{D}^2 \mathbb{D} \frac{3}{4} \mathbb{D} \gg \mathbb{D} \frac{1}{2} \tilde{N}^{\text{FLD}} \mathbb{D} \frac{3}{4} \tilde{N}^{\text{BPH}} \tilde{N}^{\text{HEL}}$ $\mathbb{D} \frac{3}{4} \tilde{N}^{\text{PAD}} \mathbb{D} \mu \mathbb{D} \frac{3}{4} \mathbb{D}^3 \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \tilde{N}^{\text{IND}} \mathbb{D}, \mathbb{D},$ $\hat{A} \ll \mathbb{D}^2 \mathbb{D} \frac{3}{4} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{FLU}} \mathbb{D} \frac{1}{4} \tilde{N}^{\text{FUE}} \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \mathbb{D}, \hat{A} \gg$	TRX-11-gw-choreography	PASS
TRX-12	$\mathbb{D}^{\text{CSI}} \mathbb{D}^{\circ} \mathbb{D} \cdot \mathbb{D} \mu \tilde{N}^{\text{PAD}} \mathbb{D} \frac{1}{2} \mathbb{D}^{\circ} \tilde{N}^{\text{SSR}}$ $\tilde{N}^{\text{BPH}} \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \tilde{N}^{\text{HOF}} \tilde{N}^{\text{HOF}} \mathbb{D}^{\circ}:$ $\tilde{N}^{\text{HEH}} \mathbb{D}$ $\mathbb{D} \mu \tilde{N}^{\text{PAD}} \mathbb{D}^{\circ} \mathbb{D} \frac{1}{2} \mathbb{D}, \mathbb{D} \mu$ $\tilde{N}^{\text{HOF}} \mathbb{D}^2 \mathbb{D} \mu \tilde{N}^{\text{BPH}} \mathbb{D} \frac{3}{4} \mathbb{D}^2 \mathbb{D} \frac{3}{4} \mathbb{D}$ $\mathbb{D} \frac{3}{4}$ $\mathbb{D} \mathbb{D}^{\circ} \tilde{N}^{\text{PAD}} \tilde{N}^{\text{HEH}} \tilde{N}^{\text{HOF}} \mathbb{D}^{\circ} \tilde{N}^{\text{HEH}}$ $\tilde{N}^{\text{BPH}} \mathbb{D} \frac{3}{4} \tilde{N}^{\text{ESR}} \mathbb{D} \mu \mathbb{D}^{\circ} \text{L4/L5}$	TRX-12-laser-light-sail	PASS

2. Досье исследований

Каждое досье цитирует критерии приёмки в точности так, как они закоммичены в `research/TRX-*/results/trx*_results.json` — проверка, измеренное значение, цель, допуск и

проверка	измерено	цель	допуск	Статус
L1_shifts_toward_radiating_primary	1	1	1.000e-12	ЗАЧТ.
Jacobi_drift_L4_orbit_beta0.05	8.882e-16	0	1.000e-10	ЗАЧТ.

TRX-02 — Đ ĐμĐ·Đ³⁄₄Đ²Đ°Đ½Ñ^{HOP}Đ½Đ³⁄₄Đμ Ñ^{SPH}Ñ^{PAD}Ñ^{FUL}Ñ^{HEL}Đ²Đ³⁄₄Đ»Đ½Đ³⁄₄Đ²Đ³⁄₄Đμ
Đ²Đ·Đ°Đ·Đ½⁄₄Đ´ĐμĐ¹Ñ^{HOP}Ñ^{SPH}Đ²Đ·Đμ (ĐSTÑ^{RI}Đ½Đ»Đ·â^{PAD}“Đ Đ³⁄₄Ñ^{HEL}, Ì^{ESA}â^{HOP}½²â^{HOP}³⁄₄-
Đ³⁄₄Đ;Ñ^{SPH}Đ·Đ°Đ°)

[illegible]

Исследование	TRX-02-three-wave-mixing
Статус	PASS
Время полного прогона	0.6 s
Code	research/TRX-02-three-wave-mixing/code/
Monograph	publications/pdf/TRX-02-three-wave-mixing_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
ManleyRow e_I13_drift	5.107e-15	0	1.000e-10	ЗАЧТ.

проверка	измерено	цель	допуск	Статус
ManleyRow e_I23_drift	1.776e-15	0	1.000e-10	ЗАЧТ.
ManleyRow e_I12diff_drift	3.775e-15	0	1.000e-10	ЗАЧТ.
Pump_perio d_repeatabi lity	4.470e-08	0	1.000e-06	ЗАЧТ.
Pump_reviv al_error	2.442e-15	0	1.000e-08	ЗАЧТ.
Pump_full_d epletion_oc curs	1	1	1.000e-12	ЗАЧТ.
SHG_tanh2 _conversion error	2.220e-16	0	1.000e-08	ЗАЧТ.

TRX-03 — $\mathbb{D}^{\text{ST}} \mathbb{D}^3 \mathbb{D} \gg \mathbb{D} \mu \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\text{NH}} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D} \cdot \tilde{\mathbb{N}}^{\text{EPH}} \tilde{\mathbb{N}}^{\text{PAD}} \tilde{\mathbb{N}}^{\text{FUE}} \tilde{\mathbb{N}}^{\text{NEL}} \tilde{\mathbb{N}}^{\text{HOP}} \mathbb{D}^3 \mathbb{D} \gg \mathbb{D} \cdot \tilde{\mathbb{N}}^{\text{EPH}} \mathbb{D}^3 \mathbb{D}^{\frac{1}{2}} \mathbb{D}^3 \mathbb{D}^2 \mathbb{D}^2$
 $\mathbb{D}^2 \mathbb{D}^3 \mathbb{D} \gg \mathbb{D}^3 \mathbb{D}^{\circ} \mathbb{D}^3 \mathbb{D}^{\frac{1}{2}} \mathbb{D}^{\frac{1}{2}} \mathbb{D}^3 \mathbb{D}^{\frac{1}{4}} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D} \cdot \mathbb{D} \mu \tilde{\mathbb{N}}^{\text{PAD}} \mathbb{D} \mu \tilde{\mathbb{N}}^{\text{HOP}} \tilde{\mathbb{N}}^{\text{HOP}} \mathbb{D} \cdot \mathbb{D}^{\frac{1}{2}} \tilde{\mathbb{N}}^{\text{NEL}} \tilde{\mathbb{N}}^{\text{PAD}} \mathbb{D}^3 \mathbb{D}^2 \mathbb{D} \cdot \mathbb{D} \cdot \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\text{SSR}} \mathbb{D} \cdot \mathbb{D} \mu \mathbb{D}^1$
 $\mathbb{D}^3 \mathbb{D}^3 \mathbb{D}^{\circ}$

[illegible]

Исследование	TRX-03-soliton-molecule
Статус	PASS
Время полного прогона	1.2 s (4.7 s with –figures)
Code	research/TRX-03-soliton-molecule/code/
Monograph	publications/pdf/TRX-03-soliton-molecule_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
pair_equilibrium_numeric_vs_analytic	5.551e-17	0	1.000e-10	ЗАЧТ.
antiphase_pi2_purely_repulsive	1	1	1.000e-12	ЗАЧТ.
molecule_final_spacings_equal	2.220e-16	0	1.000e-08	ЗАЧТ.
molecule_final_spacing_equals_s_star	2.776e-17	0	1.000e-06	ЗАЧТ.
conservative_energy_drift	1.332e-15	0	1.000e-10	ЗАЧТ.
breathing_mode_frequency	0.468825	0.468766	0.00468766	ЗАЧТ.

TRX-04 — $\text{D}\alpha\text{D}^4\text{N}^{\text{SPH}}\text{D}^3\text{D}^1\text{D}^1\text{D}^0\text{N}^{\text{SSX}}\text{D}\text{T}\text{D}, \text{D}^1\text{D}^0\text{D}^3\text{N}^{\text{HOP}}\text{N}^{\text{SPH}}\text{N}^{\text{FLU}}: \text{N}^{\text{SPH}}\text{N}^{\text{PAD}},$
 $\text{D}^0\text{D}\mu\text{N}^{\text{PAD}}\text{N}^{\text{PAD}}\text{D}^3\text{D}^2\text{N}^{\text{HOP}}\text{D}^0\text{D}, \text{N}^{\text{HEL}}\text{N}^{\text{HOP}}\text{D}^3\text{D}^2\text{D}, \text{N}^{\text{SPH}}\text{D}^3\text{D}^1\text{D}^0$
 $(\text{D}; \text{N}^{\text{PAD}}\text{D}^3\text{N}^{\text{HOP}}\text{N}^{\text{SPH}}\text{N}^{\text{PAD}}\text{D}^0\text{D}^1\text{N}^{\text{HOP}}\text{N}^{\text{SPH}}\text{D}^2\text{D}\mu\text{D}^1\text{D}^1\text{N}^{\text{FLD}}\text{D}\mu, 2\text{-D})$

DĈ[N_{FAD}Ĉ, D»D°D·DμN_{FAD}D½N_{FLG}N_{HEL} DĉN_{NBH}N_{ESA}D°D°,
 N_{FAD}D°N_{HOP}DĉN_{FAD}D¾N_{HOP}N_{SPH}N_{FAD}D°D½N_{SS3}N_{SSN}N_{HTJ}D, N_{HEL}N_{HOP}N_{SS3} N_{HOP}D¾4D²D¼DμN_NN_{FAD}D½D¾ D²
 N_{HOP}N_{FAD}DμD' Dμ N_{HOP} N_{IND}D¾4D°N_{NBH}N_{HOP}D, N_{FAD}N_{NBH}N_{SS2}N_{HTJ}DμD¹ D°DμN_{FAD}N_{FAD}D¾4D²N_{HOP}D°D¾4D¹
 D½DμD»D, D½DμD¹D½D¾4N_{HOP}N_{SPH}N_{FLU}N_{SS2} D²DμD' N_{NBH}N_{SPH} N_{HOP}DμD±N_{SS3} D°D°D°
 N_{IND}D¾4N_{SPH}D¾4D½D½D°N_{SS3} D¶D, D' D°D¾4N_{HOP}N_{SPH}N_{FLU}: D°D°D¶D' N_{FLG}D¹ DĉN_{NBH}N_{ESA}D¾4D° â_{FADCC}
 DĉN_{FAD}D¾4N_{HOP}N_{SPH}N_{FAD}D°D½N_{HOP}N_{SPH}D²DμD½D½N_{FLG}D¹ N_{HOP}D¾4D»D, N_{SPH}D¾4D½
 D½DμD»D, D½DμD¹D½D¾4D³D¾ N_{NBH}N_{FAD}D°D²D½DμD½D, N_{SS3} D' N_{FAD}N_{FU}D' D, D½D³DμN_{FAD}D°, D°
 DĉD°N_{FAD}N_{FLG} DĉN_{NBH}N_{ESA}D°D¾4D² D¾4D±D¼DμD½D, D²D°N_{SS2}N_{SPH}N_{HOP}N_{SS3}
 D°D¾4D½N_{HOP}DμN_{FAD}D²D°N_{SPH}D, D²D½N_{FLG}D¼4D, D' D²N_{NBH}N_{HEL}N_{SPH}DμD»N_{FLU}D½N_{FLG}D¼4D, N_{HOP}D, D»D°D¼D,,
 D·D½D°D° D°D¾4N_{SPH}D¾4N_{FAD}N_{FLG}N_{HEL} D·D°D' D°N_{FU}N_{SPH} D¾4N_{SPH}D½D¾4N_{HOP}D, N_{SPH}DμD»N_{FLU}D½D°N_{SS3}

$\tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D}^\circ \hat{a}_{\text{FADCC}} \tilde{N}_{\text{HOP}} \text{D}, \text{D} \frac{1}{2} \tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \text{D} \mu \text{D} \text{D} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{ESA}} \text{D}^\circ \text{D},$
 $\text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}, \tilde{N}_{\text{EPH}} \tilde{N}_{\text{SSZ}} \text{D}^3 \text{D}, \text{D}^2 \text{D}^\circ \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}} \text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{EPH}} \text{D}, \text{D}^2 \text{D}^3 \tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \text{D} \mu$
 $\text{D}^3 \tilde{N}_{\text{EPH}} \tilde{N}_{\text{EPH}} \text{D}^\circ \text{D} \gg \text{D}^\circ \text{D}, \text{D}^2 \text{D}^\circ \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}} \hat{a}_{\text{FADCC}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{RI}} \text{D}^\circ \tilde{N}_{\text{HOP}} \text{D} \text{D} \text{D}^3 \text{D} \frac{1}{2} \text{D} \mu \text{D} \frac{1}{2} \tilde{N}_{\text{ESA}} \text{D}, \text{D}^\circ \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \text{D} \frac{1}{4}$
 $\text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{IND}} \text{D}, \text{D} \gg \text{D} \mu \text{D} \frac{1}{4} \text{V}(\text{r}) = \hat{a}_{\text{HTS}}^{\text{HTS}} \text{U}\hat{\text{A}} \cdot \text{e}^{\wedge}(\hat{a}_{\text{HTS}}^{\text{HTS}} \text{r}/w). \text{D} \text{D} \text{D}, \text{D} \frac{1}{2} \tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \text{D}^1 \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \text{D}, \text{D} \text{D} \gg \text{D} \mu \tilde{N}_{\text{EPH}}$
 $\tilde{N}_{\text{FAD}} \text{D} \mu \text{D}^\circ \text{D} \gg \text{D}, \text{D} \cdot \tilde{N}_{\text{NEH}} \text{D} \mu \tilde{N}_{\text{EPH}} \text{D}^3 \text{D} \text{D} \tilde{N}_{\text{EPH}} \text{D}, \tilde{N}_{\text{ESA}} \text{D} \mu \tilde{N}_{\text{HOP}} \text{D}^\circ \tilde{N}_{\text{NEH}} \tilde{N}_{\text{SSZ}} \text{D} \gg \text{D}^\circ \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^\circ \text{D} \frac{1}{2} \text{D} \mu \text{D}^2 \tilde{N}_{\text{NEH}}$
 $\tilde{N}_{\text{SSZ}} \text{D} \mu \text{D} \frac{1}{2} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \text{D}^\circ \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{SSZ}} \text{D}^\circ \text{D}^3 \text{D} \frac{1}{2} \tilde{N}_{\text{IND}} \text{D}, \text{D}^3 \tilde{N}_{\text{NEH}} \tilde{N}_{\text{FAD}} \text{D}^\circ \tilde{N}_{\text{SSZ}} \text{D}, \tilde{N}_{\text{SSZ}}:$
 $\tilde{N}_{\text{FAD}} \text{D}^\circ \text{D}^2 \text{D} \frac{1}{2} \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^3 \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}, \text{D}^1 \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \text{D} \mu \tilde{N}_{\text{NEH}} \text{D}^3 \text{D}^3 \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \text{D}, \text{D}^\circ \text{D} \text{D} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{ESA}} \text{D}^\circ \text{D}^3 \text{D}^2$
 $\tilde{N}_{\text{HOP}} \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^3 \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}^1 \text{a} = 3 \text{D} \mu \tilde{N}_{\text{FLU}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^\circ \text{D}^3 \text{D}^2 \text{D}^2 \tilde{N}_{\text{FAD}} \text{D}^\circ \tilde{N}_{\text{HTJ}} \text{D}^\circ \text{D} \mu \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{HOP}} \text{D} \text{D} \text{D} =$
 $\text{e}^{\wedge}(\hat{a}_{\text{HTS}}^{\text{HTS}} \text{r}/2) = 0.223130160 \hat{a}_{\text{FAD}}^{\text{FAD}} \tilde{N}_{\text{RI}} \text{D}^\circ \tilde{N}_{\text{HOP}} \text{D} \text{D} \text{D}^3 \text{D} \frac{1}{2} \text{D} \mu \text{D} \frac{1}{2} \tilde{N}_{\text{ESA}} \text{D}, \text{D}^\circ \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \text{D} \frac{1}{4}$
 $\text{D}^\circ \text{D} \frac{1}{2} \text{D}^\circ \text{D} \gg \text{D}^3 \text{D}^3 \text{D} \frac{1}{4} \text{D} \frac{1}{2} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{EPH}} \text{D}^3 \text{D} \frac{1}{2} \text{D}^3 \text{D}^2 \tilde{N}_{\text{HOP}} \text{D}^\circ \text{D}^3 \text{D}^1 \text{I}_{\text{HTJ}} \hat{\text{A}}^2 = 3 \text{Gm}/\text{a} \hat{\text{A}}^3 \text{D}, \text{D} \cdot$
 $\tilde{N}_{\text{EPH}} \text{D} \mu \text{D}^3 \tilde{N}_{\text{FAD}} \text{D} \mu \text{D} \frac{1}{4} \tilde{N}_{\text{FLD}} 3.1. \text{D}^{\text{SSZ}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{HOP}} \text{D} \gg \text{D} \mu \text{D}^{\text{D}} \text{D}^3 \text{D}^2 \text{D}^\circ \text{D} \frac{1}{2} \text{D}, \text{D} \mu \text{D}^2 \text{D} \mu \tilde{N}_{\text{FAD}} \text{D}, \tilde{N}_{\text{IND}} \text{D}, \tilde{N}_{\text{SSZ}} \text{D}, \tilde{N}_{\text{FAD}} \tilde{N}_{\text{NEH}} \text{D} \mu \tilde{N}_{\text{EPH}}$
 $\tilde{N}_{\text{NEH}} \text{D}^3 \tilde{N}_{\text{FAD}} \text{D} \mu \text{D}^3 \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^\circ \tilde{N}_{\text{IND}} \text{D}, \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{HOP}} \text{D} \frac{1}{4} \text{D}^\circ \tilde{N}_{\text{HTS}} \text{D}, \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}^3 \text{D}^1 \tilde{N}_{\text{EPH}} \text{D}^3 \tilde{N}_{\text{ESA}} \text{D} \frac{1}{2} \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{SSZ}}:$
 $\tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \text{D} \mu \tilde{N}_{\text{NEH}} \text{D}^3 \text{D}^3 \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \text{D}, \text{D}^\circ \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^\circ \tilde{N}_{\text{FLU}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}}$
 $\tilde{N}_{\text{FAD}} \text{D}^\circ \text{D}^2 \text{D} \frac{1}{2} \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^3 \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}, \text{D} \frac{1}{4} \text{D}^{\text{D}} \text{D}^3 \text{D} \frac{1}{2} \text{D}^\circ \text{D} \text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{EPH}} \tilde{N}_{\text{SSZ}} \text{D} \mu \text{D} \frac{1}{2} \text{D}, \text{D},$
 $\tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{NEH}} \text{D} \text{D} \text{D}^3 \text{D} \gg \text{D} \frac{1}{2} \tilde{N}_{\text{FLD}} \tilde{N}_{\text{NEH}} \text{D}^3 \text{D}^3 \text{D} \pm \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{EPH}} \text{D}^3 \text{D}^2, \text{D}, \text{D} \cdot \text{D} \frac{1}{4} \text{D} \mu \tilde{N}_{\text{FAD}} \text{D} \mu \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}^\circ \tilde{N}_{\text{SSZ}}$
 $\tilde{N}_{\text{NEH}} \text{D}^3 \text{D} \gg \text{D}^3 \text{D}^2 \text{D}^\circ \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{HOP}} \text{D}^\circ \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{HOP}} \text{D}^3 \text{D}^2 \text{D} \text{D} \text{D}^\circ \text{D}^{\text{D}} \text{D}^\circ \text{D} \mu \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}}$
 $\text{D}^\circ \text{D} \frac{1}{2} \text{D}^\circ \text{D} \gg \text{D}, \tilde{N}_{\text{EPH}} \text{D}, \tilde{N}_{\text{ESA}} \text{D} \mu \tilde{N}_{\text{HOP}} \text{D}^\circ \text{D}, \text{D} \frac{1}{4} \text{D} \cdot \text{D}^\circ \text{D}^\circ \text{D}^3 \text{D} \frac{1}{2} \text{D}^3 \text{D} \frac{1}{4} \text{D} \frac{1}{4} \text{D}^{\text{D}} \text{D}^3 \text{D} \frac{1}{2} \tilde{N}_{\text{EPH}} \hat{\text{A}}^1 \hat{\text{A}}^1,$
 $\tilde{N}_{\text{RI}} \text{D} \frac{1}{2} \text{D} \mu \tilde{N}_{\text{FAD}} \text{D}^3 \text{D}, \tilde{N}_{\text{SSZ}} \text{D}, \text{D} \frac{1}{4} \text{D}^3 \text{D} \frac{1}{4} \text{D} \mu \text{D} \frac{1}{2} \tilde{N}_{\text{EPH}} \text{D}, \text{D} \frac{1}{4} \text{D} \text{D} \tilde{N}_{\text{NEH}} \text{D} \gg \tilde{N}_{\text{FLU}} \tilde{N}_{\text{HOP}} \text{D}^\circ \tilde{N}_{\text{HOP}} \text{D}^3 \tilde{N}_{\text{NEH}}$
 $\tilde{N}_{\text{FAD}} \text{D}^\circ \text{D} \frac{1}{2} \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}} \text{D}^{\text{D}} \text{D}^3 \text{D} \frac{1}{4} 4.7\text{e-}16 \text{D}, 2.2\text{e-}15, \text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}^3 \tilde{N}_{\text{EPH}} \text{D}, \text{D}^2 \text{D}^3 \tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D} \frac{1}{2} \text{D}^3 \text{D} \mu$
 $\tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \text{D}, \text{D}^3 \tilde{N}_{\text{ESA}} \text{D}, \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^3 \tilde{N}_{\text{FAD}} \text{D}^\circ \text{D} \cdot \text{D} \gg \text{D} \mu \tilde{N}_{\text{EPH}} \text{D}^\circ \text{D} \mu \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}} \text{D}^{\text{D}} \text{D}^3 \text{D} \frac{1}{4} 23.476, \text{D}^\circ$
 $\tilde{N}_{\text{HOP}} \text{D}, \text{D} \frac{1}{2} \tilde{N}_{\text{IND}} \text{D}^\circ \text{D} \cdot \text{D} \frac{1}{2} \text{D}^\circ \tilde{N}_{\text{SSZ}} \text{D} \pm \text{D}, \text{D} \frac{1}{2} \text{D}^\circ \tilde{N}_{\text{FAD}} \text{D} \frac{1}{2} \text{D}^\circ \tilde{N}_{\text{SSZ}} \text{D} \text{D} \text{D}^\circ \tilde{N}_{\text{FAD}} \text{D}^\circ \text{D}^3 \tilde{N}_{\text{HOP}} \tilde{N}_{\text{EPH}} \text{D}^\circ \tilde{N}_{\text{FLU}} \tilde{N}_{\text{EPH}} \tilde{N}_{\text{HOP}} \tilde{N}_{\text{SSZ}}$
 $\tilde{N}_{\text{HOP}} \text{D}^2 \tilde{N}_{\text{SSZ}} \text{D} \cdot \text{D}^\circ \text{D} \frac{1}{2} \text{D} \frac{1}{2} \text{D}^3 \text{D}^1 \tilde{N}_{\text{HOP}} \text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D} \mu \tilde{N}_{\text{SSZ}} \text{D} \mu \tilde{N}_{\text{HOP}} \tilde{N}_{\text{HOP}} \text{D}, \tilde{N}_{\text{FAD}} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{HTJ}} \text{D} \mu \text{D}^1 \text{D}^3 \tilde{N}_{\text{FAD}} \text{D} \pm \text{D}, \tilde{N}_{\text{EPH}} \text{D}^3 \text{D}^1$
 $\hat{a}_{\text{FADCC}} \text{D}^3 \text{D} \text{D} \text{D} \tilde{N}_{\text{EPH}} \text{D}, \tilde{N}_{\text{ESA}} \text{D} \mu \tilde{N}_{\text{HOP}} \text{D}^\circ \text{D}^\circ \tilde{N}_{\text{SSZ}} \tilde{N}_{\text{EPH}} \text{D}^\circ \text{D} \pm \text{D} \gg \text{D}, \tilde{N}_{\text{SSZ}} \text{D}^\circ \tilde{N}_{\text{EPH}} \tilde{N}_{\text{FAD}} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{EPH}} \text{D} \mu \text{D} \gg \tilde{N}_{\text{FLU}} \text{D} \frac{1}{2} \text{D}^3 \text{D}^3 \text{D}^3 \text{D}^3$
 $\tilde{N}_{\text{FAD}} \text{D}^\circ \tilde{N}_{\text{HOP}} \tilde{N}_{\text{HOP}} \text{D} \mu \tilde{N}_{\text{SSZ}} \text{D} \frac{1}{2} \text{D}, \tilde{N}_{\text{SSZ}} \text{D} \cdot \text{D}^\circ \text{D} \frac{1}{4} \text{D}^\circ \text{D} \frac{1}{2} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{EPH}} \text{D}^\circ \tilde{N}_{\text{SSZ}} 9/9 \text{D} \text{D} \tilde{N}_{\text{FAD}} \text{D}^3 \text{D}^2 \text{D} \mu \tilde{N}_{\text{FAD}} \text{D}^\circ \text{D}^\circ \text{D} \frac{1}{4} \text{D}, \text{PASS}.$

Исследование	TRX-04-photon-fluid
Статус	PASS
Время полного прогона	9.4 s (9.415 s recorded with –figures)
Code	research/TRX-04-photon-fluid/ code/
Monograph	publications/pdf/TRX-04-photon- fluid_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
triangle_eq uilateral_de viation	4.441e-15	0	1.000e-06	3АЧТ.
measured_r otation_rate	0.22313	0.22313	1.000e-08	3АЧТ.
energy_con	2.220e-16	0	1.000e-10	3АЧТ.

проверка	измерено	цель	допуск	Статус
conservation_rotation				
angular_momentum_conservation	8.882e-16	0	1.000e-10	ЗАЧТ.
antiphase_nano_collapse	1	1	1.000e-12	ЗАЧТ.
antiphase_expands	1	1	1.000e-12	ЗАЧТ.
energy_conservation_repulsion	3.053e-16	0	1.000e-10	ЗАЧТ.
binary_stays_bound	1	1	1.000e-12	ЗАЧТ.
binary_energy_conservation	1.771e-12	0	1.000e-10	ЗАЧТ.

TRX-05 — $\mathbb{D}^{\text{PM}}\mathbb{D}_{\xi}\tilde{\mathbb{N}}^{\text{SPH}}\mathbb{D}_{\tilde{\mathbb{N}}}^{\text{ESR}}\mathbb{D}_{\mu\tilde{\mathbb{N}}}^{\text{HOP}}\mathbb{D}^{\circ}\mathbb{D}_{\mathbb{D}}\mathbb{D}^2\mathbb{D}_{\tilde{\mathbb{N}}}\mathbb{N}^{\text{EL}}\mathbb{N}^{\text{PAD}}\mathbb{D}_{\mathbb{D}}\mathbb{D}^{\circ}\mathbb{D}_{\mathbb{D}}\mu\tilde{\mathbb{N}}^{\text{PAD}}\mathbb{D}^{\frac{1}{2}}\tilde{\mathbb{N}}^{\text{PLD}}\tilde{\mathbb{N}}^{\text{EL}}\mathbb{D}_{\xi}\tilde{\mathbb{N}}^{\text{HOB}}\tilde{\mathbb{N}}^{\text{ESR}}\mathbb{D}^{\circ}\mathbb{D}^3\mathbb{D}^2\mathbb{D}_{\mathbb{D}}\mathbb{D}^{\circ}\mathbb{D}^{\frac{1}{2}}\mathbb{D}^{\circ}\mathbb{D}^{\circ}\mathbb{D}^{\circ}\mathbb{D}^{\circ}\mathbb{D}^3\mathbb{D}_{\tilde{\mathbb{N}}}\tilde{\mathbb{N}}^{\text{SS}}\tilde{\mathbb{N}}^{\text{HOP}}\tilde{\mathbb{N}}^{\text{SPH}}\mathbb{D}^3\tilde{\mathbb{N}}^{\text{ESR}}\mathbb{D}_{\mu\tilde{\mathbb{N}}}^{\text{ESR}}\mathbb{D}^{\frac{1}{2}}\tilde{\mathbb{N}}^{\text{PLD}}\mathbb{D}^{\frac{1}{4}}\mathbb{D}_{\mathbb{D}}\mathbb{D}^2\mathbb{D}_{\tilde{\mathbb{N}}}\mathbb{N}^{\text{EL}}\tilde{\mathbb{N}}^{\text{PAD}}\tilde{\mathbb{N}}^{\text{SS}}\mathbb{D}^{\frac{1}{4}}\mathbb{D}_{\mathbb{D}}$

[illegible]

$\tilde{N}_{\text{BPH}} D^3/4 \tilde{N}_{\text{ESR}} D^{1/2} D^3/4 \tilde{N}_{\text{HOF}} \tilde{N}_{\text{BPH}} D, 3 \hat{a}_{\text{FADCC}} D^\circ D^{1/2} D^\circ D \gg D^3/4 D^3 \tilde{N}_{\text{FAD}} D \pm D, \tilde{N}_{\text{BPH}} D^\circ D \gg \tilde{N}_{\text{FLU}} D^{1/2} D^3/4 D^3 D^3/4$
 $\tilde{N}_{\text{NEH}} D^3 D \gg D^3/4 D^2 D^3/4 D^3 D^3/4 D^{1/4} D^3/4 D^{1/4} D \mu D^{1/2} \tilde{N}_{\text{BPH}} D^\circ 3 \hat{a}_{\text{INDSS}} D^{1/2} D^\circ \tilde{N}_{\text{IND}} D^3/4 \tilde{N}_{\text{BPH}} D^3/4 D^{1/2}.$

Исследование	TRX-05-optical-vortices
Статус	PASS
Время полного прогона	4.83 s (4.827 s recorded with –figures)
Code	research/TRX-05-optical-vortices/code/
Monograph	publications/pdf/TRX-05-optical-vortices_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
rotation_rate_v s_analytic	0.477465	0.477465	1.000e-08	3АЧТ.
angular_impuls e_I_conserved	1.332e-15	0	1.000e-12	3АЧТ.
kirchhoff_hami ltonian_conser ved	3.534e-16	0	1.000e-12	3АЧТ.
field_minima_tr ack_vortices	0.016125	0	0.06	3АЧТ.
total_winding_ number_is_3	3	3	1.000e-12	3АЧТ.

TRX-06 — $D_{\text{DCS}} \tilde{N}_{\text{BPH}} D^3/4 D^{1/4} D^3 D \mu D \gg D, \tilde{N}_{\text{SS}} D^\circ D^\circ D^\circ D^\circ \tilde{N}_{\text{NEH}} D \gg D^3/4 D^{1/2} D^3/4 D^2 \tilde{N}_{\text{HOF}} D^\circ D^\circ \tilde{N}_{\text{SS}} D \cdot D^\circ D$
 $'D^\circ \tilde{N}_{\text{ESR}} D^\circ \tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} \tilde{N}_{\text{FUI}} \tilde{N}_{\text{NEL}} \tilde{N}_{\text{BPH}} D \mu D \gg (D^\circ D \gg D^\circ \tilde{N}_{\text{HOF}} \tilde{N}_{\text{HOF}} D, \tilde{N}_{\text{ESR}} D \mu \tilde{N}_{\text{HOF}} D^\circ D, D^1$
 $\tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} D^\circ D \mu D^\circ \tilde{N}_{\text{BPH}} D^3/4 \tilde{N}_{\text{FAD}} D^{1/2} \tilde{N}_{\text{FLD}} D^1 D^{1/4} D \mu \tilde{N}_{\text{BPH}} D^3/4 D' D_{\text{ST}} D^3/4 D^{1/2} \tilde{N}_{\text{BPH}} D \mu \cdot D_{\text{SCI}} D^\circ \tilde{N}_{\text{FAD}} D \gg D^3/4)$

$D_{\text{DCS}} \tilde{N}_{\text{BPH}} D^3/4 D^{1/4} D^3 D \mu D \gg D, \tilde{N}_{\text{SS}} \hat{a}_{\text{FADCC}} \tilde{N}_{\text{SS}} D' \tilde{N}_{\text{FAD}} D^3/4 \tilde{N}_{\text{HOF}} D \cdot D^\circ \tilde{N}_{\text{FAD}} \tilde{N}_{\text{SS}} D' D^3/4 D^{1/4} Z = 2 D, D' D^2 D^\circ$
 $\tilde{N}_{\text{RI}} D \gg D \mu D^\circ \tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} D^3/4 D^{1/2} D^\circ \hat{a}_{\text{FADCC}} D^{1/2} D^\circ D, D^{1/4} D \mu D^{1/2} \tilde{N}_{\text{FLU}} \tilde{N}_{\text{HTS}} D^\circ \tilde{N}_{\text{SS}} \tilde{N}_{\text{IND}} D, D \cdot D, \tilde{N}_{\text{ESR}} D \mu \tilde{N}_{\text{HOF}} D^\circ D^\circ \tilde{N}_{\text{SS}}$
 $\tilde{N}_{\text{HOF}} D, \tilde{N}_{\text{HOF}} \tilde{N}_{\text{BPH}} D \mu D^{1/4} D^\circ, D^\circ D \gg D^\circ \tilde{N}_{\text{HOF}} \tilde{N}_{\text{HOF}} D, \tilde{N}_{\text{ESR}} D \mu \tilde{N}_{\text{HOF}} D^\circ D, D^1 D_{\text{Z}} \tilde{N}_{\text{FAD}} D \mu D' D \mu D \gg D^\circ D^3/4 \tilde{N}_{\text{BPH}} D^3/4 \tilde{N}_{\text{FAD}} D^3/4 D^1$
 $\tilde{N}_{\text{SS}} D^2 D \gg \tilde{N}_{\text{SS}} D \mu \tilde{N}_{\text{BPH}} \tilde{N}_{\text{HOF}} \tilde{N}_{\text{SS}} \tilde{N}_{\text{ESR}} D \mu \tilde{N}_{\text{HOF}} \tilde{N}_{\text{BPH}} D^{1/2} D^3/4 D^1 D \cdot D^\circ D' D^\circ \tilde{N}_{\text{ESR}} D \mu D^1 \tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} \tilde{N}_{\text{FUI}} \tilde{N}_{\text{NEL}} \tilde{N}_{\text{BPH}} D \mu D \gg: D$
 $'D^2 D^\circ D_{\text{Z}} \tilde{N}_{\text{FAD}} D, \tilde{N}_{\text{BPH}} \tilde{N}_{\text{SS}} D^3 D, D^2 D^\circ \tilde{N}_{\text{SS}} \tilde{N}_{\text{HTJ}} D, \tilde{N}_{\text{NEL}} D, D^3/4 D' D^{1/2} D^3/4 D^3/4 \tilde{N}_{\text{BPH}} \tilde{N}_{\text{BPH}} D^\circ D \gg D^\circ D, D^2 D^\circ \tilde{N}_{\text{SS}} \tilde{N}_{\text{SS}} \tilde{N}_{\text{HTJ}} D \mu D \mu$
 $D^\circ \tilde{N}_{\text{NEH}} D \gg D^3/4 D^{1/2} D^3/4 D^2 \tilde{N}_{\text{HOF}} D^\circ D^3/4 D \mu D^2 D \cdot D^\circ D, D^{1/4} D^3/4 D' D \mu D^1 \tilde{N}_{\text{HOF}} \tilde{N}_{\text{BPH}} D^2 D, D \mu, D^2 \tilde{N}_{\text{HOF}} D \mu D_{\text{Z}} D^3/4 \tilde{N}_{\text{FAD}} \tilde{N}_{\text{SS}} D$
 $'D^\circ D^\circ D \mu D' D, D^{1/2} D, \tilde{N}_{\text{SS}} \tilde{N}_{\text{FLD}} D_{\text{SS}} \tilde{N}_{\text{HOF}} \tilde{N}_{\text{HOF}} D \gg D \mu D' D^3/4 D^2 D^\circ D^{1/2} D, D \mu D_{\text{Z}} \tilde{N}_{\text{FAD}} D^3/4 D^3 D^3/4 D^{1/2} \tilde{N}_{\text{SS}} D \mu \tilde{N}_{\text{BPH}} D$
 $'D \mu \tilde{N}_{\text{BPH}} D \mu \tilde{N}_{\text{FAD}} D^{1/4} D, D^{1/2} D, \tilde{N}_{\text{FAD}} D^3/4 D^2 D^\circ D^{1/2} D^{1/2} \tilde{N}_{\text{FLD}} D^1 D^\circ D \gg D^\circ \tilde{N}_{\text{HOF}} \tilde{N}_{\text{HOF}} D, \tilde{N}_{\text{ESR}} D \mu \tilde{N}_{\text{HOF}} D^\circ D, D^1$
 $\tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} D^\circ D \mu D^\circ \tilde{N}_{\text{BPH}} D^3/4 \tilde{N}_{\text{FAD}} D^{1/2} \tilde{N}_{\text{FLD}} D^1 D^\circ D^{1/2} \tilde{N}_{\text{HOF}} D^\circ D^{1/4} D \pm D \gg \tilde{N}_{\text{FLU}} D_{\text{ST}} D^3/4 D^{1/2} \tilde{N}_{\text{BPH}} D \mu \cdot D_{\text{SCI}} D^\circ \tilde{N}_{\text{FAD}} D \gg D^3/4$
 $(\text{CTMC}) D, D \cdot 3200 \tilde{N}_{\text{BPH}} \tilde{N}_{\text{FAD}} D^\circ D \mu D^\circ \tilde{N}_{\text{BPH}} D^3/4 \tilde{N}_{\text{FAD}} D, D^1, D \cdot D^\circ D_{\text{Z}} \tilde{N}_{\text{NEH}} \tilde{N}_{\text{HTJ}} D \mu D^{1/2} D^{1/2} \tilde{N}_{\text{FLD}} \tilde{N}_{\text{NEL}} D^2$
 $D^\circ D^3/4 D^{1/2} \tilde{N}_{\text{IND}} D, D^3 \tilde{N}_{\text{NEH}} \tilde{N}_{\text{FAD}} D^\circ \tilde{N}_{\text{ESR}} D, D_{\text{FUI}} D^\circ D^{1/2} D^{1/2} \tilde{N}_{\text{FLU}} D \mu \hat{a}_{\text{FADCC}} \tilde{N}_{\text{FAD}} D^\circ D' D, D^\circ D \gg \tilde{N}_{\text{FLU}} D^{1/2} D^3/4$

$\tilde{N}^{\text{FAD}}_{\text{D}} \cdot \text{D}^{\frac{1}{2}} \text{D} \mu \tilde{N}^{\text{HOP}}_{\text{F}} \text{F}^{\frac{1}{2}} \text{D}^{\frac{1}{2}} \text{D}^{\circ} \tilde{N}^{\text{SS}}_{\text{S}} \tilde{N}^{\text{SS}}_{\text{S}} \text{D} \mu \text{D} \text{E} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{ES}}_{\text{S}} \text{D}^{\circ} \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}^{\circ} \text{D}^3 \text{D}, \text{D} \text{E} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D} \text{E} \tilde{N}^{\text{NEH}}_{\text{H}} \tilde{N}^{\text{HOP}}_{\text{D}} \mu$
 $R^{\text{SPH}}_{\text{FAD}} = 3 \text{D}^{\frac{1}{2}} \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{ES}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \text{D}^1 \tilde{N}^{\text{RI}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^3 \text{D} \mu \tilde{N}^{\text{SPH}}_{\text{D}}, \tilde{N}^{\text{ES}}_{\text{D}} \text{D} \mu \tilde{N}^{\text{HOP}}_{\text{D}} \text{D}^{\circ} \text{D}^{\frac{3}{4}} \text{D}^1$
 $\text{D}^{\frac{3}{4}} \text{D} \pm \text{D}^{\frac{3}{4}} \text{D} \gg \text{D}^{\frac{3}{4}} \tilde{N}^{\text{ES}}_{\text{D}} \text{D}^{\circ} \text{D} \mu, \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{IND}}_{\text{D}} \text{D}, \text{D}^{\circ} \tilde{N}^{\text{HOP}}_{\text{D}} \text{D}, \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^2 \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}^{\frac{1}{2}} \tilde{N}^{\text{FLD}}_{\text{D}} \text{D}^{\frac{1}{4}} \text{D} \text{E} \text{D}, \text{D}^{\frac{1}{2}} \text{D}^{\circ} \text{D}^{\frac{3}{4}} \text{D}^{\frac{1}{4}} \tilde{I}^{\text{NEH}}_{\text{H}} = 0.01 \text{D}, \text{D}$
 $\text{D}^{\frac{3}{4}} \text{D}^{\circ} \tilde{N}^{\text{NEH}}_{\text{H}} \text{D}^{\frac{1}{4}} \text{D} \mu \text{D}^{\frac{1}{2}} \tilde{N}^{\text{SPH}}_{\text{D}}, \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^2 \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}^{\frac{1}{2}} \tilde{N}^{\text{FLD}}_{\text{D}} \text{D}^{\frac{1}{4}} \tilde{N}^{\text{HOP}}_{\text{D}} \text{D}^{\frac{1}{4}} \tilde{N}^{\text{SS}}_{\text{S}} \text{D}^3 \tilde{N}^{\text{ES}}_{\text{D}} \text{D} \mu \text{D}^{\frac{1}{2}} \text{D}, \text{D} \mu \text{D}^{\frac{1}{4}} \hat{\mu} = 0.1 \text{D}^{\circ} \cdot \text{D} \mu.,$
 $\hat{a}^{\text{FAD}}_{\text{CCH}} \text{D}, \text{D}, \text{D} \cdot \text{D}^{\frac{1}{4}} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \tilde{N}^{\text{SS}}_{\text{S}} \text{D} \mu \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\circ} \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}^{\circ} \text{D} \gg \text{D}^{\circ} \text{D}^2 \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}, \text{D}^{\frac{3}{4}} \text{D}^{\frac{1}{2}} \text{D}, \text{D} \cdot \text{D}^{\circ} \tilde{N}^{\text{ES}}_{\text{D}} \text{D}, \text{D}, : \text{D}^{\frac{3}{4}} \text{D}^{\circ} \text{D}, \text{D}^{\frac{1}{2}}$
 $\tilde{N}^{\text{RI}}_{\text{D}} \text{D} \gg \text{D} \mu \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^{\frac{1}{2}} \text{D} \cdot \text{D}^{\circ} \tilde{N}^{\text{NEL}}_{\text{D}} \text{D}^2 \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{FLD}}_{\text{D}} \text{D}^2 \text{D}^{\circ} \text{D} \mu \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{SS}}_{\text{S}}, \text{D}^{\circ} \text{D}^2 \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^1 \tilde{N}^{\text{NEH}}_{\text{H}} \tilde{N}^{\text{NEL}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}$
 $^{\circ} \text{D}, \tilde{N}^{\text{SPH}}_{\text{D}}, \tilde{N}^{\text{NEH}}_{\text{H}} \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{SS}}_{\text{S}} \text{D}^2 \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{FAD}}_{\text{D}} \text{D} \mu \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D} \mu \text{D}^{\frac{1}{4}} 7.28 \hat{A}^{\text{EFA}}_{\text{D}} \text{D}, \text{D} \cdot \text{D} \pm \tilde{N}^{\text{FLD}}_{\text{D}} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{ES}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \text{D}^1$
 $\tilde{N}^{\text{RI}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^3 \text{D}, \text{D}, \hat{a}^{\text{FAD}}_{\text{CCH}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D} \text{E} \text{D} \mu \tilde{N}^{\text{ES}}_{\text{D}} \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{FAD}}_{\text{D}} \tilde{N}^{\text{FUI}}_{\text{D}} \tilde{N}^{\text{NEL}}_{\text{D}} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D} \mu \text{D}, \tilde{N}^{\text{FLU}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \text{D}^3 \text{D}^{\frac{3}{4}}$
 $\text{D} \text{E} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D} \mu \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{HOP}}_{\text{D}} \text{D}^{\circ} \tilde{N}^{\text{RI}}_{\text{D}} \text{D}^{\frac{1}{2}} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^3 \text{D}, \text{D}, \tilde{N}^{\text{ES}}_{\text{D}} \text{D} \mu \tilde{N}^{\text{FAD}}_{\text{D}} \text{D} \mu \text{D} \cdot \text{D}^{\frac{3}{4}} \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\circ} \text{D} \gg \text{D}^{\circ} \text{D}, \text{D}^2 \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}, \text{D} \mu$
 $\text{e} \hat{a}^{\text{HOP}}_{\text{D}} \gg \hat{a}^{\text{FAD}}_{\text{D}} \text{e} \hat{a}^{\text{HOP}}_{\text{D}} \gg, \text{D} \text{E} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\circ} \tilde{N}^{\text{SPH}}_{\text{D}}, \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{SPH}}_{\text{D}}, \text{D}^{\circ} \text{D}^{\circ} \text{D}^{\circ} \text{D}^2 \text{D}^{\frac{3}{4}} \text{D}^1 \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \text{D}^3 \text{D}^{\frac{3}{4}} \tilde{N}^{\text{NEH}}_{\text{H}} \tilde{N}^{\text{NEL}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^{\circ}$
 $\tilde{N}^{\text{HOP}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D} \text{E} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{SPH}}_{\text{D}} \text{D}^{\circ} \text{D}^2 \text{D} \gg \tilde{N}^{\text{SS}}_{\text{S}} \text{D} \mu \tilde{N}^{\text{SPH}}_{\text{D}} \tilde{N}^{\text{HOP}}_{\text{D}} \tilde{N}^{\text{SS}}_{\text{S}} \tilde{N}^{\text{HOP}}_{\text{D}} \text{D} \text{E} \text{D}^{\frac{3}{4}} \tilde{N}^{\text{FAD}}_{\text{D}} \text{D}^{\frac{3}{4}} \text{D}^3 \text{D}^{\frac{3}{4}} \text{D}^2 \tilde{N}^{\text{FLD}}_{\text{D}} \text{D}^{\frac{1}{4}} \text{D} \cdot \text{D}^{\circ} \text{D}^{\circ} \text{D}^{\frac{3}{4}} \text{D}^{\frac{1}{2}} \text{D}^{\frac{3}{4}} \text{D}^{\frac{1}{4}}$
 $\text{D}^{\text{FUZ}}_{\text{D}} \text{D}^{\circ} \text{D}^{\frac{1}{2}} \text{D}^{\frac{1}{2}} \tilde{N}^{\text{FLU}}_{\text{D}} \text{D} \mu \text{P}_{\text{DE}} \hat{a}^{\text{HTS}}_{\text{SSC}} \text{E}^{\wedge} 1.056, \text{D}^$

Критерии приёмки

[illegible]

Исследование	TRX-07-efimov
Статус	PASS
Время полного прогона	5.77 s (5.771 s recorded with <code>–figures</code>)
Code	research/TRX-07-efimov/code/
Monograph	publications/pdf/TRX-07-efimov_RU.pdf

проверка	измерено	цель	допуск	Статус
s0 transcen	1.00624	1.00625	1.000e-05	ЗАЧТ.

проверка	измерено	цель	допуск	Статус
dental_root				
efimov_leng th_ratio	22.6944	22.7	0.05	ЗАЧТ.
efimov_ener gy_ratio	515.035	515.03	0.05	ЗАЧТ.
spectrum_al l_negative	1	1	1.000e-12	ЗАЧТ.
ladder_ratio _E0_over_E 1	515.477	515.035	180.262	ЗАЧТ.
ladder_ratio _E1_over_E 2	514.932	515.035	180.262	ЗАЧТ.

TRX-08 — ĐƚŃ_{FAD}Đ, Đ»Đ°Đ-ĐμŃ_{FAD}Đ½Đ¾-Đ¾Ń_{NEL}Đ»Đ°ĐŦĐ°Ń_{FUL}Đ½Đ½Ń_{FLD}Ń_{NEL}Đ, Đ¾Đ½Đ°Đ²Đ»Đ, Đ½ĐμĐ¹Đ½Đ¾Đ¹Đ»Đ¾Đ²Ń_{NH}Ń_{HTS}Đ°ĐμĐ_{FPC}Đ°Ń_{NH}Đ»Ń_{SSS}: Đ½Đ°Ń_{NH}Ń_{SPH}Đ¾Đ»Ń_{FUL}Đ½Ń_{FLD}Đ¹Ń_{SPH}Ń_{FAD}ĐμŃ_{NH}Đ³Đ¾Đ»Ń_{FUL}Đ½ĐĐ°ĐĐ_{ST}Đ°Đ³Ń_{FAD}Đ°Đ½ĐŦĐ°

$\mathbb{D}(\mathbb{N}^{\mathbb{FAD}} \mathbb{D}, \mathbb{D} \gg \mathbb{D} \cdot \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{N}^{\mathbb{HEL}} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D} \mathbb{I} \mathbb{D} \tilde{\mathbb{N}}^{\mathbb{FUL}} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{1}{2} \tilde{\mathbb{N}}^{\mathbb{FLD}} \tilde{\mathbb{N}}^{\mathbb{HEL}} \mathbb{D}, \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D}^{\circ} \mathbb{D}^2$
 $\mathbb{D}_{\mathbb{L}} \mathbb{D} \frac{3}{4} \mathbb{D}_{\mathbb{L}} \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D} \mu \mathbb{N}^{\mathbb{ESR}} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{D}^1 \mathbb{D}_{\mathbb{L}} \mathbb{D} \gg \mathbb{D} \frac{3}{4} \mathbb{N}^{\mathbb{HOF}} \mathbb{D}^{\circ} \mathbb{D} \frac{3}{4} \mathbb{N}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}, \mathbb{D} \gg \mathbb{D}, \mathbb{D} \frac{1}{2} \mathbb{D} \mu \mathbb{D}^1 \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{D}^1$
 $\mathbb{D} \gg \mathbb{D} \frac{3}{4} \mathbb{D}^2 \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{N}^{\mathbb{HTS}} \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D} \mathbb{F} \mathbb{C} \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{NHEH}} \gg \tilde{\mathbb{N}}^{\mathbb{SSZ}} \mathbb{D} \frac{3}{4} \mathbb{D} \pm \mathbb{N}^{\mathbb{FAD}} \mathbb{D}^{\circ} \mathbb{D} \cdot \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{N}^{\mathbb{SSZ}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{1}{2} \mathbb{D}^{\circ} \mathbb{N}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{3}{4} \mathbb{D} \gg \tilde{\mathbb{N}}^{\mathbb{FLU}} \mathbb{D} \frac{1}{2} \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{N}^{\mathbb{SSZ}}$
 $\mathbb{D} \cdot \mathbb{D}^{\circ} \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{ESR}} \tilde{\mathbb{N}}^{\mathbb{NHEH}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{N}^{\mathbb{FAD}} \mathbb{N}^{\mathbb{FUL}} \mathbb{N}^{\mathbb{HEL}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \mu \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D}^2 \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{D} \frac{1}{4} \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D} \frac{1}{2} \mathbb{N}^{\mathbb{FLD}} \mathbb{D}^1$
 $\mathbb{D}^3 \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D} \frac{1}{4} \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D}, \tilde{\mathbb{N}}^{\mathbb{ESR}} \mathbb{D} \mu \mathbb{N}^{\mathbb{HOF}} \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D}^1 \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D}^{\circ} \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{ESR}} \mathbb{D}^{\circ} \mathbb{N}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{1}{2} \mathbb{N}^{\mathbb{FLD}} \mathbb{D}^1 \mathbb{D}_{\mathbb{L}} \mathbb{N}^{\mathbb{HOF}} \mathbb{D} \mu \mathbb{D}^2 \mathbb{D}$
 $\mathbb{D} \frac{3}{4} \mathbb{D}_{\mathbb{L}} \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \mu \mathbb{D} \frac{1}{2} \tilde{\mathbb{N}}^{\mathbb{ESR}} \mathbb{D}, \mathbb{D}^{\circ} \mathbb{D} \gg \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{D}^{\circ} \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D} \mathbb{I} \mathbb{D}, \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D} \mu \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}, \tilde{\mathbb{N}}^{\mathbb{HEL}} \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{D} \gg \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{D}^2 \tilde{\mathbb{N}}^{\mathbb{HOF}} \mathbb{D}^{\circ} \mathbb{D} \frac{3}{4} \mathbb{D} \mu$
 $\mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{BPH}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}^{\circ} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D} \frac{1}{2} \mathbb{D}, \mathbb{D} \mu \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}^{\circ} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D} \mu \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}^{\circ} \mathbb{D}$
 $\mathbb{D} \frac{3}{4} \mathbb{D}_{\mathbb{L}} \mathbb{D} \gg \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D} \frac{3}{4} \mathbb{D}^2 \tilde{\mathbb{N}}^{\mathbb{HOF}} \mathbb{D}^{\circ} \mathbb{D} \frac{3}{4} \mathbb{D} \mu \mathbb{D} \frac{3}{4} \mathbb{N}^{\mathbb{HEL}} \mathbb{D} \gg \mathbb{D}^{\circ} \mathbb{D} \mathbb{I} \mathbb{D}^{\circ} \mathbb{D} \mu \mathbb{D} \frac{1}{2} \mathbb{D}, \mathbb{D} \mu \hat{\mathbb{a}}^{\mathbb{FADCC}} \mathbb{D}^2 \mathbb{D}^1 \mathbb{D} \frac{3}{4} \mathbb{D}^1 \mathbb{D}^{\circ} \mathbb{D} \mu \mathbb{D} \gg \mathbb{D},$
 $\mathbb{D} \gg \mathbb{D}, \mathbb{D} \frac{1}{2} \mathbb{D} \mu \mathbb{D}^1 \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{D}^3 \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{BPH}} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D} \mu \mathbb{D} \frac{1}{2} \mathbb{D}, \tilde{\mathbb{N}}^{\mathbb{SSZ}} \hat{\mathbb{a}}^{\mathbb{HTSPUZ}} \mathbb{I}^3 \mathbb{D} \hat{\mathbb{a}}^{\mathbb{FADCC}} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D}^{\circ} \mathbb{N}^{\mathbb{HOF}} \mathbb{N}^{\mathbb{HOF}} \mathbb{D} \mu \mathbb{D}, \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D} \mu \tilde{\mathbb{N}}^{\mathbb{BPH}}$
 $\tilde{\mathbb{N}}^{\mathbb{RI}} \mathbb{D} \frac{1}{2} \mathbb{D} \mu \mathbb{N}^{\mathbb{FAD}} \mathbb{D}^3 \mathbb{D}, \tilde{\mathbb{N}}^{\mathbb{SSZ}} \mathbb{D}_{\mathbb{L}} \mathbb{D} \frac{3}{4} \mathbb{D}^{\circ} \mathbb{D}^{\circ} \mathbb{D}, \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \tilde{\mathbb{N}}^{\mathbb{FLD}} \mathbb{D} \frac{1}{2} \mathbb{D} \mu \mathbb{D}^{\circ} \mathbb{N}^{\mathbb{FAD}} \mathbb{D}, \tilde{\mathbb{N}}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D}^{\circ} \mathbb{D} \gg \mathbb{D} \gg \mathbb{D}, \mathbb{D} \cdot \tilde{\mathbb{N}}^{\mathbb{NHEH}} \mathbb{N}^{\mathbb{SSZ}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{N}^{\mathbb{HOF}} \mathbb{N}^{\mathbb{SSZ}}$
 $\mathbb{D}^2 \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D}^{\circ} \mathbb{D}^2 \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{N}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{1}{2} \mathbb{D}, \mathbb{D}^1 \tilde{\mathbb{N}}^{\mathbb{BPH}} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D} \mu \mathbb{N}^{\mathbb{NHEH}} \mathbb{D}^3 \mathbb{D} \frac{3}{4} \mathbb{D} \gg \tilde{\mathbb{N}}^{\mathbb{FLU}} \mathbb{D} \frac{1}{2} \mathbb{D}, \mathbb{D}^{\circ}$
 $\mathbb{D}^{\mathbb{ESI}} \mathbb{D}^{\circ} \mathbb{D}^3 \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D}^{\circ} \mathbb{D} \frac{1}{2} \mathbb{D} \mathbb{I} \mathbb{D}^{\circ} \tilde{\mathbb{N}}^{\mathbb{HOF}} \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{HOF}} \tilde{\mathbb{N}}^{\mathbb{BPH}} \mathbb{D} \frac{3}{4} \tilde{\mathbb{N}}^{\mathbb{FAD}} \mathbb{D} \frac{3}{4} \mathbb{D} \frac{1}{2} \mathbb{D} \frac{3}{4} \mathbb{D}^1 \mathbb{a} = (3\hat{\mathbb{I}}^{\circ} / \mathbb{I}^{\mathbb{HTJ}} \hat{\mathbb{a}}^{\mathbb{BPHFAD}} \hat{\mathbb{A}}^2)^{(1/3)} = 3^{(1/3)} =$
 $1.4422495703. \mathbb{D}^{\mathbb{SSZ}} \tilde{\mathbb{N}}^{\mathbb{HOF}} \mathbb{N}^{\mathbb{HOF}} \mathbb{D} \gg \mathbb{D} \mu \mathbb{D}^{\circ} \mathbb{D} \frac{3}{4} \mathbb{D}^2 \mathbb{D}^{\circ} \mathbb{D$

$\mathcal{D}_2 \mathcal{N}^{\text{SSR}} \mathcal{N}^{\text{EPH}} \mathcal{D}_3 \mathcal{D}' \mathcal{D}_\mu \mathcal{N}^{\text{HOP}} \mathcal{N}^{\text{SSR}} \mathcal{N}^{\text{EPH}} \mathcal{D}_3 \mathcal{D}_\mu \mathcal{D}' \mathcal{D}_3 \mathcal{D}_{1/2} \mathcal{D}_3 \mathcal{N}^{\text{SSR}} \mathcal{D}^2 \mathcal{N}^{\text{FAD}} \mathcal{D}_\mu \mathcal{D}_{1/4} \mathcal{D}_\mu \mathcal{D}_{1/2} \mathcal{D}_3 \mathcal{N}^{\text{HOP}} \mathcal{D}' \mathcal{N}^{\text{FAD}} \mathcal{D}_\mu \mathcal{D}^1 \mathcal{N}^{\text{IND}} \mathcal{D}_{3/4} \mathcal{D}_{1/4}$
 $\mathcal{N}^{\text{RI}} \mathcal{D}_{1/2} \mathcal{D}_\mu \mathcal{N}^{\text{FAD}} \mathcal{D}^3 \mathcal{D}_3 \mathcal{D}_3 8.9\text{e-}16 \mathcal{N}^{\text{FAD}} \mathcal{CCH} \mathcal{D}_3 \mathcal{D}_{3/4} \mathcal{D}_{1/2} \mathcal{D}_{1/2} \mathcal{D}_{3/4} \mathcal{D} \mathcal{D} \mathcal{D}_{3/4} \mathcal{D}^2 \mathcal{N}^{\text{HBF}} \mathcal{N}^{\text{HTS}} \mathcal{D}_\mu \mathcal{N}^{\text{ESR}} \mathcal{D}_{1/2} \mathcal{N}^{\text{FLD}} \mathcal{D}^1$
 $\mathcal{D} \pm \mathcal{D} \mathcal{D} \mathcal{D}_3 \mathcal{D} \mathcal{D}_{1/2} \mathcal{D}_\mu \mathcal{N}^{\text{SSR}} \mathcal{D} \mathcal{D} \mathcal{D}^0 \mathcal{D}^3 \mathcal{N}^{\text{FAD}} \mathcal{D}^0 \mathcal{D}_{1/2} \mathcal{D} \mathcal{D}_\mu \mathcal{D}^2 \mathcal{D}^0 \mathcal{D}_{3/4} \mathcal{N}^{\text{EPH}} \mathcal{D}_{1/2} \mathcal{D}_{3/4} \mathcal{N}^{\text{HOP}} \mathcal{D}_3 \mathcal{N}^{\text{EPH}} \mathcal{D}_\mu \mathcal{D} \mathcal{D} \mathcal{N}^{\text{FLD}} \mathcal{D}_{1/2} \mathcal{D}_{3/4} \mathcal{D}^3 \mathcal{D}_{3/4}$
 $\mathcal{N}^{\text{FAD}} \mathcal{D}^0 \mathcal{D}^2 \mathcal{D}_{1/2} \mathcal{D}_{3/4} \mathcal{D}^2 \mathcal{D}_\mu \mathcal{N}^{\text{HOP}} \mathcal{D}_3 \mathcal{N}^{\text{SSR}} \mathcal{D} \mathcal{D} \mathcal{D}_\mu \mathcal{D} \mathcal{D}^0 \mathcal{N}^{\text{HTS}} \mathcal{D}_\mu \mathcal{D}^3 \mathcal{D}_{3/4} \mathcal{D}^2 \mathcal{D}_{3/4} \mathcal{N}^{\text{HOP}} \mathcal{D}_{1/2} \mathcal{D}_{3/4} \mathcal{D}^2 \mathcal{D}_\mu \mathcal{N}^{\text{EPH}} \mathcal{D}_\mu \mathcal{D}_{3/4} \mathcal{N}^{\text{FAD}} \mathcal{D}_\mu \mathcal{D}_{1/4} \mathcal{N}^{\text{FLD}}$
3.1 TRIVORTEX, $\mathcal{D}_3 \mathcal{D}_{1/2} \mathcal{D}^0 \mathcal{D}_3 \mathcal{D}_{1/4} \mathcal{D}_\mu \mathcal{D}_{1/2} \mathcal{N}^{\text{FLD}} \mathcal{N}^{\text{HTS}} \mathcal{D}_3 \mathcal{D}^1 \mathcal{D}_3 \mathcal{D}_{3/4} \mathcal{D}_{1/2} \mathcal{D}_{1/2} \mathcal{D}_{3/4}$ -
 $\mathcal{D}^0 \mathcal{N}^{\text{FAD}} \mathcal{D}_3 \mathcal{N}^{\text{HOP}} \mathcal{N}^{\text{EPH}} \mathcal{D}^0 \mathcal{D} \mathcal{D} \mathcal{D} \mathcal{D}_3 \mathcal{N}^{\text{ESR}} \mathcal{D}_\mu \mathcal{N}^{\text{HOP}} \mathcal{D}^0 \mathcal{D}_3 \mathcal{D}^1 \mathcal{D}^0 \mathcal{D}^2 \mathcal{D}^0 \mathcal{D}_{1/2} \mathcal{N}^{\text{EPH}} \mathcal{D}_{3/4} \mathcal{D}^2 \mathcal{N}^{\text{FLD}} \mathcal{D}^1 \mathcal{N}^{\text{HOP}} \mathcal{D}_3 \mathcal{D}_{1/4} \mathcal{N}^{\text{HBF}} \mathcal{D} \mathcal{N}^{\text{SSR}} \mathcal{N}^{\text{EPH}} \mathcal{D}_{3/4} \mathcal{N}^{\text{FAD}}$.

Исследование	TRX-08-ion-trap
Статус	PASS
Время полного прогона	6.79 s (6.794 s recorded with –figures)
Code	research/TRX-08-ion-trap/code/
Monograph	publications/pdf/TRX-08-ion-trap_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
global_minimum_reached	0	0	1.000e-09	3АЧТ.
crystallization_energy_released	1	1	1.000e-12	3АЧТ.
crystal_side_equality_analytic	1.44225	1.44225	1.000e-08	3АЧТ.
crystal_equilateral	0	0	1.000e-08	3АЧТ.
crystal_angles_120deg	4.441e-16	0	1.000e-06	3АЧТ.
zero_rotation_mode	1	1	1.000e-12	3АЧТ.
two_com_modes_at_omega0	1	1	1.000e-12	3АЧТ.
breathing_mode_sqrt3	1	1	1.000e-12	3АЧТ.
static_equilibrium_holds	1.388e-15	0	1.000e-09	3АЧТ.
conservative_energy_conserved	8.882e-16	0	1.000e-10	3АЧТ.

$$\begin{aligned} & \mathbb{D}_{\text{SCI}}\mathbb{D}_\circ\mathring{\mathbb{N}}_{\text{HOF}}\mathbb{N}_{\text{HOF}}\mathbb{D},\mathring{\mathbb{N}}_{\text{ESA}}\mathbb{D}\mu\mathbb{N}_{\text{HOF}}\mathbb{D}^\circ\mathbb{D}^1,\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{ESA}}\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{ESA}}\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}-\mathbb{D}^2\mathring{\mathbb{N}}_{\text{HEL}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^2\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1,\mathring{\mathbb{N}}_{\text{SSZ}}\mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{FAD}}\mathring{\mathbb{N}}_{\text{FLU}} \\ & \mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^3\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathbb{D}^1_4\mathbb{D}_{\frac{1}{4}}\mathbb{N}_{\text{PL}}\text{TRIVORTEX: }\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D},\mathbb{D}^2\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{HEL}}\mathring{\mathbb{N}}_{\text{FAD}}\mathring{\mathbb{N}}_{\text{SSZ}}\mathbb{D}_{\text{SCI}}\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{FAD}}\mathring{\mathbb{N}}_{\text{HEL}}\mathbb{D}^3\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{IND}}\mathbb{D}^\circ,\mathring{\mathbb{N}}_{\text{HOF}} \\ & \mathring{\mathbb{N}}_{\text{ESA}}\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathring{\mathbb{N}}_{\text{HBH}}\mathring{\mathbb{D}}\gg\mathring{\mathbb{N}}_{\text{SSZ}}\mathring{\mathbb{N}}_{\text{SSA}}\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{SSZ}}\mathbb{D}^1_4\mathring{\mathbb{D}},\hat{\mathbb{I}}_{\text{STS}}^1,\hat{\mathbb{I}}_{\text{TS}}^2,\hat{\mathbb{I}}_{\text{TS}}^3\mathbb{D},\mathring{\mathbb{N}}_{\text{HBH}}\mathbb{D}^3\mathring{\mathbb{D}}\gg\mathbb{D}_{\frac{3}{4}}\mathbb{D}^2\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1\mathbb{D},\mathbb{D}^1_4\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{HBH}}\mathring{\mathbb{D}}\gg\mathring{\mathbb{N}}_{\text{FLU}}\mathring{\mathbb{N}}_{\text{HOF}}\text{I=} \\ & \hat{\mathbb{I}}\hat{\mathbb{f}}_{\text{STS}}|\mathbf{r}|^{\hat{\mathbb{A}}^2}\hat{\mathfrak{a}}_{\text{FADCCH}}\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{D}},\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}_{\frac{3}{4}}\mathbb{D}_{\zeta}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}\gg\mathbb{D}_{\frac{3}{4}}\mathbb{D}^3\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{ESA}}\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{D}}^\circ\mathbb{D}_{\frac{3}{4}}\mathbb{D}^3\mathring{\mathbb{D}}_{\frac{3}{4}} \\ & \mathbb{D},\mathbb{D}^{1/2}\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}\mu\mathbb{D}^3\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathring{\mathbb{D}}\gg\mathbb{D}^\circ\mathbb{D}\S\mathbb{D}^\circ\mathbb{D}_{\zeta}\mathring{\mathbb{D}}\gg\mathring{\mathbb{N}}_{\text{FLD}}\mathbb{D}^3\mathring{\mathbb{D}},\mathbb{D}^{1/2}\mathbb{D}^\circ\text{C_Ch}\mathbb{D},\mathbb{D}\cdot\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}\mu\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^{1/4}\mathring{\mathbb{N}}_{\text{FLD}}\text{3.1.} \\ & \mathbb{D}\S\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{D}}\gg\mathbb{D}\mu\mathbb{D}^{1/2}\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^2\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^{1/2}\mathring{\mathbb{N}}_{\text{FLD}}\mathbb{D}^\circ\mathbb{D}^2\mathbb{D}^\circ\mathbb{D}^\circ\mathbb{D}^\circ\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^{1/2}\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{ESA}}\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{D}}^\circ\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{HEL}} \\ & \mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}\mathbb{D},\mathbb{D}^1_4\mathbb{D}^\circ:\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathbb{D}^2\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^{1/2}\mathbb{D}^{1/2}\mathring{\mathbb{D}},\mathbb{D}^1\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{HBH}}\mathbb{D}^3\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}\gg\mathring{\mathbb{N}}_{\text{FLU}}\mathbb{D}^{1/2}\mathring{\mathbb{D}},\mathbb{D}^\circ \\ & \mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}'\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}\cdot\mathbb{D}^{1/2}\mathbb{D}^\circ\mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathbb{D}^2\mathring{\mathbb{N}}_{\text{FLD}}\mathring{\mathbb{N}}_{\text{HEL}}\mathbb{D}^2\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{HEL}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^1\mathbb{D}^2\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathring{\mathbb{N}}_{\text{HTJ}}\mathbb{D}^\circ\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{N}}_{\text{SSZ}} \\ & \mathbb{D}\mathbb{Q}\mathring{\mathbb{N}}_{\text{FUe}}\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{N}}_{\text{BPH}}\mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{N}}_{\text{HBH}}\mathbb{D}^3\mathring{\mathbb{D}}\gg\mathbb{D}_{\frac{3}{4}}\mathbb{D}^2\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1\mathring{\mathbb{N}}_{\text{HOF}}\mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{HOF}}\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{N}}_{\text{FLU}}\mathring{\mathbb{N}}_{\text{SSZ}}[\hat{\mathbb{I}}]_{\text{HTJ}}=3\hat{\mathbb{I}}_{\text{STS}}^2/(2\hat{\mathbb{I}}_{\text{FAD}}^2\hat{\mathbb{A}}^2)= \\ & 0.477464829\left(\mathbb{D}^2\mathring{\mathbb{D}},\mathring{\mathbb{N}}_{\text{HEL}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^2\mathbb{D}_{\frac{3}{4}}\mathbb{D}\mu\mathring{\mathbb{D}}\gg\mathbb{D}^\circ\mathbb{D}^3\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathbb{D}^{1/2}\mathbb{D}\mathbb{D}\mu\mathbb{D}^2\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathring{\mathbb{N}}_{\text{HTS}}\mathbb{D}\mu\mathbb{D}^{1/2}\mathring{\mathbb{D}},\mathbb{D}\mu\right),\mathbb{D}^\circ \\ & \mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathbb{D}\cdot\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}\cdot\mathbb{D}^{1/2}\mathbb{D}^\circ\mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathbb{D}^2\mathbb{D}^\circ\mathring{\mathbb{N}}_{\text{SSZ}}\mathring{\mathbb{N}}_{\text{BPH}}\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1\mathbb{D}^\circ\mathbb{D}^\circ(1,1,\hat{\mathfrak{a}}_{\text{HTSFUE}}1),\mathbb{D}\cdot\mathbb{D}^\circ\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{HBH}}\mathring{\mathbb{N}}_{\text{HTJ}}\mathbb{D}\mu\mathbb{D}^{1/2}\mathbb{D}^{1/2}\mathbb{D}^\circ\mathring{\mathbb{N}}_{\text{SSZ}} \\ & \mathbb{D},\mathbb{D}\cdot\mathbb{D}_{\zeta}\mathring{\mathbb{N}}_{\text{FAD}}\mathring{\mathbb{N}}_{\text{SSZ}}\mathbb{D}^1_4\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{N}}_{\text{HBH}}\mathbb{D}^3\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}\gg\mathring{\mathbb{N}}_{\text{FLU}}\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}^\circ\mathbb{D}^2\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathring{\mathbb{D}}+\mathbb{D}\mu\mathring{\mathbb{D}}'\mathring{\mathbb{N}}_{\text{FAD}}\mathbb{D}\mu\mathbb{D}^{1/2}\mathbb{D}^{1/2}\mathbb{D}_{\frac{3}{4}}\mathbb{D}^1 \\ & \mathbb{D}^\circ\mathbb{D}_{\frac{3}{4}}\mathbb{D}^{1/2}\mathring{\mathbb{N}}_{\text{IND}}\mathring{\mathbb{D}},\mathbb{D}^3\mathring{\mathbb{N}}_{\text{HBH}}\mathring{\mathbb{N}}_{\text$$

Критерии приёмки

TRIVORTEX Research Program · v1.0.0 · 16 / 20

проверка	измерено	цель	допуск	Статус
linear_impulse _conserved	7.216e-16	0	1.000e-12	ЗАЧТ.
aref_collapse_c onditions_hold	1.388e-17	0	1.000e-12	ЗАЧТ.
mixed_sign_inv ariants_conser ved	1.310e-14	0	1.000e-11	ЗАЧТ.
mixed_sign_sh ape_evolves	1	1	1.000e-12	ЗАЧТ.

TRX-10 — $\text{D}_{\text{SCI}}\text{D}^3_4\text{D}\rangle\text{D}_{\mu}\text{D}\pm\text{D}^{\circ}\text{D}\frac{1}{2}\text{D}_{\tilde{\text{N}}^{\text{SS}}}\text{D}_{\text{SCI}}\text{D}^3_4\text{D}\cdot\text{D}^{\circ}\text{D}_{\hat{\text{A}}^{\text{PAD}}}\text{D}_{\text{SCI}}\text{D}_{\text{D}}\text{D}^3_4\text{D}^2\text{D}^{\circ}\text{D}^2\text{D}_{\text{D}}\text{D}_{\mu}\tilde{\text{N}}^{\text{PAD}}\text{D}^{\circ}\tilde{\text{N}}^{\text{PAD}}\text{D}_{\tilde{\text{N}}^{\text{SS}}}\text{D}_{\mu}\tilde{\text{N}}^{\text{HOF}}\text{D}^{\circ}\text{D}_{\tilde{\text{N}}^{\text{HEL}}}\tilde{\text{N}}^{\text{EPH}}\tilde{\text{N}}^{\text{PAD}}\text{D}^3_4\text{D}^1\text{D}^1\frac{1}{2}\tilde{\text{N}}^{\text{FLD}}\tilde{\text{N}}^{\text{HEL}}\tilde{\text{N}}^{\text{HOF}}\text{D}_{\tilde{\text{N}}^{\text{HOF}}}\tilde{\text{N}}^{\text{EPH}}\text{D}_{\mu}\text{D}\frac{1}{4}\text{D}^{\circ}\tilde{\text{N}}^{\text{HEL}}$

[illegible]

Исследование	TRX-10-kozai-lidov
Статус	PASS
Время полного прогона	64.6 s
Code	research/TRX-10-kozai-lidov/

publications/pdf/TRX-10-kozai-
lidov_RU.pdf

проверка	измерено	цель	допуск	Статус
e_max_i0_6 0	0.763752	0.763763	0.002	ЗАЧТ.
e_max_i0_7 0	0.897229	0.897239	0.002	ЗАЧТ.
kl_period_h alves_with_ m3	0.493173	0.5	0.03	ЗАЧТ.

[illegible]

Исследование	TRX-11-gw-choreography
Статус	PASS
Время полного прогона	8.17 s
Code	research/TRX-11-gw-choreography/code/
Monograph	publications/pdf/TRX-11-gw-choreography_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
period_closure_error	1.232e-08	0	5.000e-08	3АЧТ.
period_in_expected_range	1	1	1.000e-12	3АЧТ.
energy_conserved	3.331e-15	0	1.000e-12	3АЧТ.
angular_momentum_is_zero	1.443e-15	0	1.000e-09	3АЧТ.
luminosity_finite_positive	1	1	1.000e-12	3АЧТ.
dominant_harmonic_on_comb	0.00149963	0	0.05	3АЧТ.

TRX-12 — $\Delta_{\text{CSI}} \Delta^{\circ} \Delta \cdot \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{1}{2}} \Delta^{\circ} \tilde{N} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{HOP}} \Delta^{\circ} : \tilde{N} \Delta_{\text{NEH}} \Delta^{\circ} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^{\frac{1}{2}} \Delta \cdot \Delta_{\mu} \tilde{N} \Delta_{\text{HOP}} \Delta^2 \Delta_{\mu} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta^2 \Delta^{\frac{3}{4}} \Delta^3 \Delta^{\frac{3}{4}} \Delta_{\text{Z}} \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{NEH}} \Delta_{\text{HOP}} \Delta^{\circ} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{ESA}} \Delta_{\mu} \Delta^{\circ} \text{L4/L5}$

$\Delta_{\text{I}} \Delta^2 \Delta_{\mu} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta^2 \Delta^{\frac{3}{4}} \Delta^1 \Delta_{\text{Z}} \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{HOP}} \Delta^2 \Delta^{\frac{3}{4}} \Delta^3 \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{ESA}} \Delta_{\mu} \Delta^{\frac{1}{2}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^1 \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{NEH}} \Delta^3 \Delta^{\frac{3}{4}} \Delta^2 \Delta^{\frac{3}{4}} \Delta^1 \Delta \cdot \Delta^{\circ} \Delta^{\circ} \Delta^{\circ} \tilde{N} \Delta_{\text{ESA}} \Delta_{\mu} \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{PUL}} \tilde{N} \Delta_{\text{HEL}} \tilde{N} \Delta_{\text{BPH}} \Delta_{\mu} \Delta \gg \Delta_{\text{BPH}} \Delta_{\mu} \Delta^{\frac{1}{4}} \Delta \gg \tilde{N} \Delta_{\text{SS3}} \Delta_{\text{PAD}} \Delta^{\circ} \Delta_{\text{CSI}} \tilde{N} \Delta_{\text{NEH}} \Delta^{\frac{1}{2}} \Delta^{\circ},$
 $\tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta \gg \Delta^{\circ} \Delta^{\circ} \Delta_{\mu} \Delta^{\frac{1}{4}} \tilde{N} \Delta_{\text{FLD}} \Delta^1 \Delta \gg \Delta^{\circ} \Delta \cdot \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{FLD}} \Delta^{\frac{1}{4}} \Delta_{\text{Z}} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{ESA}} \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{4}}$
 $\Delta^{\frac{1}{4}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{HTJ}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{FLU}} \tilde{N} \Delta_{\text{SS2}} \Delta^1 \Delta_{\text{ST}} \Delta_{\text{PU2}} \tilde{N} \Delta_{\text{BPH}} \cdot \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{SS3}} \Delta^3 \Delta^{\circ} a_L = 2P/(\text{cm})$
 $\Delta^{\frac{1}{2}} \Delta^{\circ} \Delta_{\text{Z}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^2 \Delta \gg \tilde{N} \Delta_{\text{SS3}} \Delta_{\mu} \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{SS3}} \Delta \cdot \Delta^{\circ} \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta^1 \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{4}}$
 $\Delta_{\text{Z}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{3}{4}} \Delta_{\text{Z}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{SS3}} \Delta_{\text{FLU}} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{2}} \Delta^{\circ} \Delta \gg \tilde{N} \Delta_{\text{FLU}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^1 \Delta^{\circ} \Delta_{\text{Z}} \tilde{N} \Delta_{\text{IND}} \tilde{N} \Delta_{\text{IND}} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta_{\mu} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{SS3}} \Delta_{\text{FLU}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^3 \Delta^{\frac{3}{4}}$
 $\tilde{N} \Delta_{\text{NEH}} \Delta_{\text{Z}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^2 \Delta \gg \Delta_{\mu} \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{SS3}} \Delta_{\text{Z}} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{ESA}} \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{4}} \Delta \cdot \Delta^{\frac{3}{4}} \Delta^3 \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{ESA}} \Delta_{\mu} \Delta^{\frac{1}{2}} \Delta^{\circ}$
 $\tilde{N} \Delta_{\text{IND}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{2}} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{FLD}} \Delta^{\frac{1}{4}} \Delta_{\text{Z}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta \gg \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{4}} a_{\text{max}} = 0.2443$
 $(\Delta \pm \Delta_{\mu} \Delta \cdot \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta \cdot \Delta^{\frac{1}{4}} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{FLD}} \tilde{N} \Delta_{\text{HEL}} \Delta_{\mu} \Delta^{\circ} \Delta \cdot \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{SS3}}) \cdot \Delta_{\text{SS3}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{HOP}} \Delta \gg \Delta_{\mu} \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta^2 \Delta^{\circ} \Delta^{\frac{1}{2}} \Delta \cdot \Delta_{\mu} \Delta$
 $\Delta^{\circ} \Delta_{\mu} \Delta^{\frac{1}{4}} \Delta^{\frac{3}{4}} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{PAD}} \Delta \cdot \tilde{N} \Delta_{\text{PAD}} \tilde{N} \Delta_{\text{NEH}} \Delta_{\mu} \tilde{N} \Delta_{\text{BPH}} \Delta^{\circ} \Delta^2 \Delta_{\mu} \Delta \pm \Delta^{\circ} \Delta \cdot \Delta^{\frac{3}{4}} \Delta^2 \tilde{N} \Delta_{\text{FLD}} \Delta_{\mu} \Delta^{\frac{3}{4}} \Delta_{\text{Z}} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \tilde{N} \Delta_{\text{SS3}} \Delta \cdot \Delta,$
 $\hat{A} \ll \Delta \gg \Delta^{\circ} \Delta \cdot \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^1 \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{FLD}} \hat{A} \gg \hat{a}_{\text{PAD}} \cdot \tilde{N} \Delta_{\text{NEH}} \Delta_{\text{HOP}} \Delta^{\circ} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta^{\frac{1}{2}} \Delta \cdot \Delta_{\mu}$
 $\Delta^{\circ} \Delta_{\text{Z}} \Delta_{\text{Z}} \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \tilde{N} \Delta_{\text{BPH}} \Delta^{\circ} \tilde{N} \Delta_{\text{NEH}} \Delta \gg \Delta \cdot \Delta^{\frac{1}{2}} \Delta_{\mu} \Delta^1 \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \Delta^1 \tilde{N} \Delta_{\text{ESA}} \Delta \cdot \Delta^2 \Delta^{\frac{3}{4}} \Delta^1, \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}}$
 $\Delta^{\frac{1}{2}} \Delta_{\mu} \Delta^1 \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta \gg \tilde{N} \Delta_{\text{FLU}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \Delta_{\mu} \Delta^1 \tilde{N} \Delta_{\text{IND}} \tilde{N} \Delta_{\text{NEH}} \tilde{N} \Delta_{\text{SS2}} \tilde{N} \Delta_{\text{HTJ}} \Delta_{\mu} \Delta^1 \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{ESA}} \Delta^{\circ} \Delta \cdot \text{L4} \tilde{N} \Delta_{\text{HOP}}$
 $\tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{ESA}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{FLU}} \tilde{N} \Delta_{\text{SS2}} 7.5e-07 \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^1 \Delta^{\frac{1}{4}} \Delta \cdot \Delta^{\frac{1}{2}} \Delta^{\circ} \Delta \gg \tilde{N} \Delta_{\text{FLU}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^3 \Delta^{\frac{3}{4}}$
 $\Delta_{\text{Z}} \Delta^{\frac{3}{4}} \Delta \gg \Delta^{\frac{3}{4}} \Delta^{\circ} \Delta_{\mu} \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{SS3}} \Delta \cdot \Delta_{\text{Z}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{3}{4}} \Delta^{\circ} \Delta^{\circ} \tilde{N} \Delta_{\text{ESA}} \Delta^{\circ} \tilde{N} \Delta_{\text{NEH}} \Delta \cdot \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{BPH}} \Delta_{\mu} \Delta^3 \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta \gg \Delta^{\circ} \Delta^{\circ} \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta \pm \Delta,$
 $\Delta^2 \Delta^{\frac{1}{2}} \Delta \cdot \tilde{N} \Delta_{\text{BPH}} \tilde{N} \Delta_{\text{SS3}} \Delta^3 \Delta^{\frac{3}{4}} \Delta^1 \Delta^2 \Delta^{\circ} \Delta^{\frac{3}{4}} \Delta \gg \tilde{N} \Delta_{\text{FLU}} \tilde{N} \Delta_{\text{HOP}} \Delta^{\circ} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{PAD}} \Delta^{\frac{3}{4}} \tilde{N} \Delta_{\text{HOP}} \tilde{N} \Delta_{\text{BPH}} \Delta \cdot \Delta^{\circ} \Delta \gg \tilde{N} \Delta_{\text{Z}} \Delta_{\mu} \tilde{N} \Delta_{\text{PAD}} \Delta_{\mu} \tilde{N} \Delta_{\text{HEL}} \Delta^{\frac{3}{4}} \Delta^1 \Delta^{\circ}$
 $\Delta^{\frac{1}{4}} \Delta_{\mu} \Delta^{\circ} \Delta_{\text{NEH}} \Delta \cdot \Delta^{\frac{1}{2}} \Delta^2 \Delta^{\circ} \tilde{N} \Delta_{\text{PAD}} \Delta \cdot \Delta^{\circ} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{BPH}} \Delta^{\frac{1}{2}} \tilde{N} \Delta_{\text{FLD}} \Delta^{\frac{1}{4}} \Delta \cdot \Delta^{\frac{1}{4}} \Delta^{\frac{1}{2}} \Delta^{\frac{3}{4}} \Delta^3 \Delta^{\frac{3}{4}} \Delta^{\frac{3}{4}} \Delta \pm \tilde{N} \Delta_{\text{PAD}} \Delta^{\circ} \Delta \cdot \Delta \cdot \tilde{N} \Delta_{\text{SS3}} \Delta^1 \Delta \cdot,$

Статус	PASS
Время полного прогона	41.4 s
Code	research/TRX-12-laser-light-sail/ code/
Monograph	publications/pdf/TRX-12-laser- light-sail_RU.pdf

Критерии приёмки

проверка	измерено	цель	допуск	Статус
controlled_l 4_bound	8.763e-05	0	2.000e-04	ЗАЧТ.
control_sat uration_frac tion	0	0	0.05	ЗАЧТ.
free_drift_n ever_conver ges	1	1	1.000e-12	ЗАЧТ.
jacobi_pum ped_monoto nically	1	1	1.000e-12	ЗАЧТ.
power_table _consistent	1	1	1.000e-12	ЗАЧТ.

4. Воспроизведение

smoke: all twelve acceptance suites in seconds (what CI runs)

for d in research/TRX-*/code/*.py; **do** python3 "\$d" --smoke; **done**

full: every study, full statistics, JSON protocols

for d in research/TRX-*/code/*.py; **do** python3 "\$d"; **done**

ultra-resolution figure sets (300 dpi)

for d in research/TRX-*/code/*.py; **do** python3 "\$d" --figures; **done**

3. Дисциплина верификации — Каждое исследование — детерминированный скрипт (только numpy/scipy), который выполняет свою физику, сравнивает измеренные числа с опубликованными или зарегистрированными целями и пишет JSON-протокол с полным снимком параметров. Режим --smoke прогоняет ту же логику приёмки за секунды и встроен в CI; полный режим и режим --figures заново генерируют набор графиков 300 dpi. Исследование проходит только когда зелёны все проверки — та же дисциплина, которую основная лестница верификации V1–V4 применяет к теореме 3.1 вихревого документа.